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FORTRAN VIID: LICENSED RESTRICTED RIGHTS AS STATED IN LICENSE  ****************, SEE DOCUMENTATION PACKAGE, 04-101M99.
  1  0000001          PROGRAM  CNS3D                                                                                    1
  2                C      BY: Y. S. CHEN        ON: 8/13/1985                                                          2
  3                C                                                                                                   3
  4                C*****[ CURVILINEAR N-S CODE FOR 3-D INCOMPRESSIBLE FLOWS ]**********                               4
  5                C                                                                                                   5
  6  0000061          COMMON                                                                                          6
  7                1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),                                           7
  8                2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                                                    8
  9                3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                                         9
 10                1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                                           10
 11                1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                                                   11
 12                1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),                                 12
 13                2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),                                      13
 14                3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),                                      14
 15                4 TALW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO                                       15
 16                1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),                                    16
 17                2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),                                        17
 18                3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)                                         18
 19  0000061          COMMON                                                                                          19
 20                1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),                                       20
 21                2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                                                       21
 22                3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                                                       22
 23                3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                                                        23
 24                1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,                                      24
 25                2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT                                   25
 26  0000061          COMMON                                                                                          26
 27                1/TTRAN/TXXE(21,18,10),TXXW(21,18,10),TYYN(21,18,10),                                             27
 28                2   TYYS(21,18,10),TZZT(21,18,10),TZZB(21,18,10),                                                 28
 29                3   TYXE(21,18,10),TYXW(21,18,10),TYZT(21,18,10),                                                 29
 30                4   TYZS(21,18,10),TXYN(21,18,10),TXYS(21,18,10),                                                 30
 31                5   TXZT(21,18,10),TXZB(21,18,10),TZXE(21,18,10),                                                 31
 32                6   TZXW(21,18,10),TZYN(21,18,10),TZYS(21,18,10)                                                  32
 33                1/UNSTDY/UO(21,18,10),VO(21,18,10),WO(21,18,10),DKO(21,18,10),                                    33
 34                2   DEO(21,18,10),DENO(21,18,10),TMO(21,18,10)                                                    34
 35  0000061          LOGICAL INSOU,INSOV,INSOP,INSOK,INSOE,INPRO,INSOW,INSOT                                         35
 36                C*****[ INPUT DATA GUIDE ]*************************************************                       36
 37                C                                                                                                 37
 38                C  NLIMT : MAXIMUM NO. OF ITERATIONS LIMIT                                                        38
 39                C                                                                                                 39
 40                C  IG    = 1 : LAMINAR                                                                            40
 41                C           2 : TURBULENT (K-E MODEL)                                                            41
 42                C                                                                                                 42
 43                C  ISWP  : NO. OF SWEEPS FOR SOLVING THE P' EQUATION (PP).                                        43
 44                C                                                                                                 44
 45                C  ITT   : TOTAL NO. OF TIME STEPS.                                                               45
 46                C                                                                                                 46
 47                C  ALU,ALV,ALW,ALP,ALK,ALE,ALVIS,ALC : UNDER-RELAXATION FACTORS                                    47
 48                C                                                                                                 48
 49                C  RENL  : REFERENCE REYNOLDS NUMBER.                                                             49
 50                C                                                                                                 50
 51                C  DTT   : TIME STEP FOR UNSTEADY PROBLEMS.                                                       51
 52                C                                                                                                 52
 53                C*****************************************************************                               53
 54                C-----INPUT DATA (PROBLEM CONTROL SETTING)                                                         54
 55  0000061          READ(5,100) NLIMT,IG,ISWP,ITT                                                                   55        1
 56  0000301          READ(5,200) ALU,ALV,ALW,ALP,ALK,ALE,ALVIS,ALC                                                   56        2
 57  00006CI          READ(5,200) RENL,DTT                                                                            57        3
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 58                C-----CONSTANTS                                                                                  58
 59  000090I             EREXT=1.E-3                                                                               59        4
 60  00009CI             ISWL=7                                                                                    60        5
 61  0000A4I             ISWV=7                                                                                    61        6
 62  0000ACI             ISWW=7                                                                                    62        7
 63  0000B4I             ISWK=5                                                                                    63        8
 64  0000BCI             ISWE=5                                                                                    64        9
 65  0000C4I             DENIN=1.0                                                                                 65       10
 66  0000D0I             VISC=1./RENL                                                                              66       11
 67  0000E2I             SIGL=1.0                                                                                  67       12
 68  0000EEI             SIGK=1.0                                                                                  68       13
 69  0000FAI             SIGE=1.3                                                                                  69       14
 70  000106I             CMU=0.09                                                                                  70       15
 71  000112I             C1=1.43                                                                                   71       16
 72  00011EI             C2=1.92                                                                                   72       17
 73  00012AI             E=9.01069                                                                                 73       18
 74  000136I             CK=0.4                                                                                    74       19
 75  000142I             PI=3.141592654                                                                            75       20
 76  00014EI             RINLM=1.E30                                                                               76       21
 77  00015AI             SMNLM=1.E-30                                                                              77       22
 78  000166I             CMU1=CMU**0.25                                                                            78       23
 79  00017EI             CMU2=CMU**0.75                                                                            79       24
 80  000196I             INSCU=.TRUE.                                                                              80       25
 81  00019EI             INSCV=.TRUE.                                                                              81       26
 82  0001A6I             INSCP=.TRUE.                                                                              82       27
 83  0001AEI             INSCK=.TRUE.                                                                              83       28
 84  0001B6I             INSCE=.TRUE.                                                                              84       29
 85  0001BEI             INPRO=.TRUE.                                                                              85       30
 86  0001C6I             INSCW=.TRUE.                                                                              86       31
 87  0001CEI             INSCT=.FALSE.                                                                             87       32
 88  0001D6I             IF(IG .EQ. 2) GO TO 10                                                                    88       33
 89  0001ECI             INSCK=.FALSE.                                                                             89       35
 90  0001F4I             INSCE=.FALSE.                                                                             90       36
 91  0001FCI             INPRO=.FALSE.                                                                             91       37
 92  000204I           10  CONTINUE                                                                                92       38
 93                C*****[READ IN INITIAL FLOW FIELDS FROM RESTART FILE (LU = 8)]********                           93
 94  000204I             READ(8,100) L,M,N,L1,L2,M1,M2,N1,N2                                                       94       39
 95  000244I             LO=L+1                                                                                    95       40
 96  000252I             MO=M+1                                                                                    96       41
 97  000260I             NO=N+1                                                                                    97       42
 98  00026EI             LT=L-1                                                                                    98       43
 99  00027CI             MT=M-1                                                                                    99       44
100  00028AI             NT=N-1                                                                                   100       45
101                C-----INITIALIZE VARIABLES                                                                     101
102  000298I             CALL INIT                                                                                102       46
103                C-----RESTART FILE                                                                             103
104  0002A0I             DO 50 K=1,N                                                                              104       47
105  0002B4I             DO 50 I=1,L                                                                              105       48
106  0002C8I             READ(8,400)                                                                              106       49
107  0002E4I             DO 50 J=1,M                                                                              107       50
108  0002F8I             READ(8,500) X(I,J,K),Y(I,J,K),Z(I,J,K),U(I,J,K),V(I,J,K),                               108       51
109                     1 W(I,J,K),P(I,J,K),TM(I,J,K),DK(I,J,K),DE(I,J,K),                                       109
110                     2 VISETM,DENTM                                                                           110
111  000588I             UO(I,J,K)=U(I,J,K)                                                                       111       52
112  000604I             VO(I,J,K)=V(I,J,K)                                                                       112       53
113  000650I             WO(I,J,K)=W(I,J,K)                                                                       113       54
114  00069CI             TMO(I,J,K)=TM(I,J,K)                                                                     114       55
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115  0006E8I          DKO(I,J,K)=DK(I,J,K)                                                    115      56
116  000734I          DEO(I,J,K)=DE(I,J,K)                                                    116      57
117  000780I          DENO(I,J,K)=DEN(I,J,K)                                                  117      58
118  0007CCI       50 CONTINUE                                                                118      59
119               C*************************************************************************  119
120               C-----GET BOUNDARY CONTROL PARAMETERS                                       120
121  000814I          CALL DIRCOS                                                             121      60
122               C-----SET BOUNDARY TURBULENCE QUANTITIES TO ZERO                            122
123  00081CI          DO 121 I=1,L                                                            123      61
124  000830I          DO 121 J=1,M                                                            124      62
125  000844I          DO 121 K=1,N                                                            125      63
126  000858I          IF(MC(I,J,K) .NE. 0) GO TO 122                                          126      64
127  00088EI          GO TO 121                                                               127      66
128  0008A4I       122 DK(I,J,K)=0.0                                                          128      67
129  0008C0I          DE(I,J,K)=0.0                                                           129      68
130  0008ECI          U(I,J,K)=0.0                                                            130      69
131  000918I          V(I,J,K)=0.0                                                            131      70
132  000944I          W(I,J,K)=0.0                                                            132      71
133  000970I       121 CONTINUE                                                               133      72
134               C-----CALCULATE GRID TRANSFORMATION COEFFICIENTS                            134
135  000988I          CALL TRANF                                                              135      73
136               C-----TURBULENT VISCOSITY                                                   136
137  0009C0I          IF(INPRO) CALL NEWVIS                                                   137      74
138               C-----CALCULATE INLET MASS FLOW RATE                                        138
139  0009D4I          FLOWIN=0.0                                                              139      76
140  0009E0I          I=1                                                                     140      77
141  0009E8I          DO 45 J=2,M                                                             141      78
142  0009FCI          DO 45 K=2,N                                                             142      79
143  000A10I          UC=(U(I,J,K)+U(I,J-1,K)+U(I,J,K-1)+U(I,J-1,K-1))*0.25                   143      80
144  000AC0I          DENC=(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K-1)+DEN(I,J-1,K-1))*0.25         144      81
145  000B70I          P1=(X(I,J,K)+X(I,J,K-1)-X(I,J-1,K)-X(I,J-1,K-1))*0.5                    145      82
146  000C20I          P2=(Y(I,J,K)+Y(I,J,K-1)-Y(I,J-1,K)-Y(I,J-1,K-1))*0.5                    146      83
147  000CD0I          P3=(Z(I,J,K)+Z(I,J,K-1)-Z(I,J-1,K)-Z(I,J-1,K-1))*0.5                    147      84
148  000D80I          Q1=(X(I,J,K)+X(I,J-1,K)-X(I,J,K-1)-X(I,J-1,K-1))*0.5                    148      85
149  000E30I          Q2=(Y(I,J,K)+Y(I,J-1,K)-Y(I,J,K-1)-Y(I,J-1,K-1))*0.5                    149      86
150  000EE0I          Q3=(Z(I,J,K)+Z(I,J-1,K)-Z(I,J,K-1)-Z(I,J-1,K-1))*0.5                    150      87
151  000F90I          AREA=SQRT(P1*P1+P2*P2+P3*P3)*SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                     151      88
152  00102CI          FLOWIN=FLOWIN+DENC*AREA*UC                                              152      89
153  00104AI       45 CONTINUE                                                                153      90
154  00107AI          ITO=1                                                                   154      91
155               C-----TRANSIENT PROCESS                                                     155
156  001032I        2 CONTINUE                                                                156      92
157  001082I          CALL SYMOUT(3,1,2,L,2,M,2,N)                                            157      93
158  00100CI          ITER=1                                                                  158      94
159               C-----SOLUTION PROCEDURES START                                             159
160  0010E4I        1 CONTINUE                                                                160      95
161  0010E4I          CALL SYMOUT(1,1,2,LT,2,MT,2,NT)                                         161      96
162  001140I          IF(INSOU) CALL SOLVEQ(1,ISWU,ALU,SIGU,ERRU,U,UC)                        162      97
163  001184I          IF(INSOV) CALL SOLVEQ(2,ISWV,ALV,SIGU,ERRV,V,VO)                        163      99
164  0011C8I          IF(INSOW) CALL SOLVEQ(3,ISWW,ALW,SIGU,ERRW,W,WO)                        164     101
165  00120CI          IF(INSOT) CALL SOLVEQ(4,ISWW,ALW,SIGU,ERRW,TM,TMO)                      165     103
166  001250I          IF(INSOK) CALL SOLVEQ(5,ISWK,ALK,SIGK,ERRK,DK,DKO)                      166     105
167  001294I          IF(INSOE) CALL SOLVEQ(6,ISWE,ALE,SIGE,ERRE,DE,DEO)                      167     107
168  0012D8I          IF(INSOP) CALL SOLVEQ(0,ISWP,ALP,SIGU,ERRM,PP,PP)                       168     109
169  00131CI          IF(INPRO) CALL NEWVIS                                                   169     111
170               C-----CONVERGENCE CHECK                                                     170
171  001330I          WRITE(6,300) ITER,ERRU,ERRV,ERRW,ERRM,ERRK,ERRE,U(7,2,6)                171     113
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ORIGINAL PAGE IS
OF POOR QUALITY

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172  00136CI          ERRMAX=ERRM+ERRU+ERRV+ERRW                                                    172      114
173  00138AI          IF(ITER .GE. 20 .AND. ERRMAX .GT. 1.E03) GO TO 99                             173      115
174  0013B2I          IF(ITER .GE. NLIMT .OR. ERRMAX .LE. EREXT) GO TO 99                           174      117
175  0013DCI          ITER=ITER+1                                                                   175      119
176  0013EAI          GO TO 1                                                                       176      120
177                C-----PRINT OUT SOLUTIONS                                                       177
178  0013EEI       99 CONTINUE                                                                      178      121
179  0013EEI          WRITE(7,100) L,M,N,L1,L2,M1,M2,N1,N2                                          179      122
180  00142CI          DO 901 K=1,N                                                                  180      123
181  001440I          DO 901 I=1,L                                                                  181      124
182  001454I          WRITE(7,400)                                                                  182      125
183  001470I          DO 902 J=1,M                                                                  183      126
184  001484I          XV=X(I,J,K)                                                                   184      127
185  0014B0I          YV=Y(I,J,K)                                                                   185      128
186  0014DCI          ZV=Z(I,J,K)                                                                   186      129
187  0015C8I          WRITE(7,500) XV,YV,ZV,U(I,J,K),V(I,J,K),W(I,J,K),P(I,J,K),                    187      130
188                1           TM(I,J,K),DK(I,J,K),DE(I,J,K),VISE(I,J,K),DEN(I,J,K)                188
189  0016E8I      902 CONTINUE                                                                     189      131
190  001700I      901 CONTINUE                                                                     190      132
191  001730I          TIMT=DTT*ITC                                                                  191      133
192  001744I          WRITE(7,300) ITC,TIMT                                                         192      134
193  001768I          IF(ITC .GE. ITT .OR. DTT .EQ. C.0) GO TO 999                                  193      135
194  001792I          ITC=ITC+1                                                                     194      137
195  0017A0I          GO TO 2                                                                       195      138
196  0017A4I      999 CONTINUE                                                                     196      139
197  0017A4I      100 FORMAT(9I5)                                                                  197
198  0017B0I      200 FORMAT(11F7.4)                                                               198
199  0017BEI      300 FORMAT(1X,I5,7E10.2)                                                         199
200  0017C0I      400 FORMAT(//)                                                                   200
201  0017DAI      500 FORMAT(3F8.4,3E12.4,2X,6E11.4)                                               201
202  0017F6I          STOP                                                                          202      140
203  0017FEI          END                                                                           203      141

NO  ERRORS:F7D  R05-01.0C  MAINPROG    CNS3C     02/21/86  09:45:56  TABLE SPACE:  11 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 181 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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 1  000000I         SUBROUTINE DIRCCS                                                  204
 2  000004I         COMMON                                                             205
 3                 1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),   206
 4                 2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),        207
 5                 3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),        208
 6                 4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO         209
 7                 1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),         210
 8                 2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                        211
 9                 3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                        212
10                 3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                         213
11                 1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LC,MO,ISWU,ISWV,ISWP,ISWK,ISWE,       214
12                 2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT    215
13           C-----SET DOMAIN BLOCKAGE CONTROL PARAMETER                               216
14           C-----SCALAR BLOCKAGE :   MC(I,J,K)=1                                     217
15           C-----PRESSURE BLOCKAGE : MC(I,J,K)=2                                     218
16  000004I       DO 10 I=1,LO                                                         219        1
17  000018I       DO 10 J=1,MO                                                         220        2
18  00002CI       DO 10 K=1,NO                                                         221        3
19  000040I       MC(I,J,K)=0                                                          222        4
20  000068I       IF(J.EQ.1.OR.J.EQ.M.OR.K.EQ.N) MC(I,J,K)=1                           223        5
21  0000C4I       IF(I.GE.L1.AND.I.LE.L2.AND.J.GE.M1.AND.J.LE.M2.AND.                  224        7
22                1   K.GE.N1.AND.K.LE.N2) MC(I,J,K)=1                                225
23  000158I       IF(I.GT.L1.AND.I.LE.L2.AND.J.GT.M1.AND.J.LE.M2.AND.                  226        9
24                1   K.GT.N1.AND.K.LE.N2) MC(I,J,K)=2                                227
25           C-----ADD BLOCKAGES AS NEEDED HERE                                        228
26  0001ECI     10  CONTINUE                                                         229       11
27           C-----CALCULATE BOUNDARY GRID SIZES AND ORIENTATIONS                      230
28  000234I       III=1                                                              231       12
29  00023CI       DO 30 I=2,LT                                                       232       13
30  000250I       DO 30 J=2,MT                                                       233       14
31  000264I       DO 30 K=2,NT                                                       234       15
32  000278I       IF(MC(I,J,K) .NE. 0) GO TO 30                                      235       16
33  0002AEI       MCT=MC(I+1,J,K)+MC(I-1,J,K)+MC(I,J+1,K)+MC(I,J-1,K)+               236       18
34                1    MC(I,J,K+1)+MC(I,J,K-1)                                       237
35  0003ACI       IF(MCT .EQ. 0) GO TO 30                                            238       19
36  0003C2I       IF(MC(I,J+1,K) .EQ. 0) GO TO 2                                     239       21
37           C-----NORTH                                                             240
38  0003FCI       IBC(III)=I                                                         241       23
39  000410I       JBC(III)=J                                                         242       24
40  000424I       KBC(III)=K                                                         243       25
41  000438I       IITY(III)=1                                                        244       26
42  000448I       I1=I+1                                                             245       27
43  000456I       I2=I-1                                                             246       28
44  000464I       K1=K+1                                                             247       29
45  000472I       K2=K-1                                                             248       30
46  000480I       IF(I .EQ. L1) I2=I                                                 249       31
47  00049EI       IF(I .EQ. L2) I1=I                                                 250       33
48  00048CI       IF(K .EQ. N1) K2=K                                                 251       35
49  0004DAI       IF(K .EQ. N2) K1=K                                                 252       37
50  0004F8I       J1=J-1                                                             253       39
51  000506I       J2=J-2                                                             254       40
52  000514I       P1=(Y(I1,J,K1)-Y(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))-                255       41
53                1   (Z(I1,J,K1)-Z(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))                 256
54  000650I       P2=(Z(I1,J,K1)-Z(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))-                 257       42
55                1   (X(I1,J,K1)-X(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))                 258
56  00078CI       P3=(X(I1,J,K1)-X(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))-                 259       43
57                1   (Y(I1,J,K1)-Y(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))                 260
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 58  0008C8I          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                    261      44
 59  000912I          P1=P1/PQ                                                      262      45
 60  000924I          P2=P2/PQ                                                      263      46
 61  000936I          P3=P3/PQ                                                      264      47
 62  000948I          R1=(1.-P1**2)                                                 265      48
 63  000964I          R2=(1.-P2**2)                                                 266      49
 64  000980I          R3=(1.-P3**2)                                                 267      50
 65  00099CI          SINX(III)=SQRT(R1)                                            268      51
 66  0009C2I          SINY(III)=SQRT(R2)                                            269      52
 67  0009E6I          SINZ(III)=SQRT(R3)                                            270      53
 68  000A0AI          Q1=X(I,J,K)-X(I,J1,K)                                         271      54
 69  000A5CI          Q2=Y(I,J,K)-Y(I,J1,K)                                         272      55
 70  000AAEI          Q3=Z(I,J,K)-Z(I,J1,K)                                         273      56
 71  000B00I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                     274      57
 72  000B7EI          CC=1.0                                                        275      58
 73  000B8AI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                    276      59
 74  000BD6I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                            277      60
 75  000C18I          YN(III)=BB*ABS(COTH)                                          278      61
 76  000C44I          Q1=X(I,J,K)-X(I,J2,K)                                         279      62
 77  000C96I          Q2=Y(I,J,K)-Y(I,J2,K)                                         280      63
 78  000CE8I          Q3=Z(I,J,K)-Z(I,J2,K)                                         281      64
 79  000D3AI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                    282      65
 80  000D86I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                     283      66
 81  000E02I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                            284      67
 82  000E44I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                           285      68
 83  000E84I          IJLC(I,J,K)=III                                               286      69
 84  000E80I          III=III+1                                                     287      70
 85  000E8EI      2   CONTINUE                                                      288      71
 86  000E8EI          IF(MC(I,J-1,K) .EQ. 0) GO TO 3                                289      72
 87               C-----SOUTH                                                      290
 88  000EF8I          IBC(III)=I                                                    291      74
 89  000F0CI          JBC(III)=J                                                    292      75
 90  000F20I          KBC(III)=K                                                    293      76
 91  000F34I          IITY(III)=2                                                   294      77
 92  000F44I          I1=I+1                                                        295      78
 93  000F52I          I2=I-1                                                        296      79
 94  000F60I          K1=K+1                                                        297      80
 95  000F6EI          K2=K-1                                                        298      81
 96  000F7CI          J1=J+1                                                        299      82
 97  000F8AI          J2=J+2                                                        300      83
 98  000F98I          P1=(Y(I1,J,K1)-Y(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))-           301      84
 99                  1   (Z(I1,J,K1)-Z(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))           302
100  0010D4I          P2=(Z(I1,J,K1)-Z(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))-           303      85
101                 1   (X(I1,J,K1)-X(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))           304
102  001210I          P3=(X(I1,J,K1)-X(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))-           305      86
103                 1   (Y(I1,J,K1)-Y(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))           306
104  00134CI          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                    307      87
105  001396I          P1=P1/PQ                                                      308      88
106  0013A8I          P2=P2/PQ                                                      309      89
107  0013BAI          P3=P3/PQ                                                      310      90
108  0013CCI          R1=(1.-P1**2)                                                 311      91
109  0013E8I          R2=(1.-P2**2)                                                 312      92
110  001404I          R3=(1.-P3**2)                                                 313      93
111  001420I          SINX(III)=SQRT(R1)                                            314      94
112  001446I          SINY(III)=SQRT(R2)                                            315      95
113  00146AI          SINZ(III)=SQRT(R3)                                            316      96
114  00148EI          Q1=X(I,J,K)-X(I,J1,K)                                         317      97
```

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115  0014E0I          Q2=Y(I,J,K)-Y(I,J1,K)                                          318      98
116  001532I          Q3=Z(I,J,K)-Z(I,J1,K)                                          319      99
117  001584I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      320     100
118  001602I          CC=1.0                                                         321     101
119  00160EI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     322     102
120  00165AI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             323     103
121  00169CI          YN(III)=BB*ABS(COTH)                                           324     104
122  0016C8I          Q1=X(I,J,K)-X(I,J2,K)                                          325     105
123  00171AI          Q2=Y(I,J,K)-Y(I,J2,K)                                          326     106
124  00176CI          Q3=Z(I,J,K)-Z(I,J2,K)                                          327     107
125  0017BEI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     328     108
126  001804I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      329     109
127  001886I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             330     110
128  0018C8I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                            331     111
129  001903I          IJLC(I,J,K)=III                                                332     112
130  001934I          III=III+1                                                      333     113
131  001942I      3   CONTINUE                                                       334     114
132  001942I          IF(MC(I+1,J,K) .EQ. 0) GO TO 4                                 335     115
133               C-----EAST                                                         336
134  001978I          IBC(III)=I                                                     337     117
135  001990I          JBC(III)=J                                                     338     118
136  0019A4I          KBC(III)=K                                                     339     119
137  0019B8I          IITY(III)=3                                                    340     120
138  0019C8I          J1=J+1                                                         341     121
139  0019D6I          J2=J-1                                                         342     122
140  0019E4I          K1=K+1                                                         343     123
141  0019F2I          K2=K-1                                                         344     124
142  001A00I          IF(J .EQ. M1) J2=J                                             345     125
143  001A1EI          IF(J .EQ. M2) J1=J                                             346     127
144  001A3CI          IF(K .EQ. N1) K2=K                                             347     129
145  001A5AI          IF(K .EQ. N2) K1=K                                             348     131
146  001A78I          I1=I-1                                                         349     133
147  001A86I          I2=I-2                                                         350     134
148  001A94I          P1=(Y(I,J1,K1)-Y(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))-             351     135
149                  1   (Z(I,J1,K1)-Z(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))             352
150  001B00I          P2=(Z(I,J1,K1)-Z(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))-             353     136
151                  1   (X(I,J1,K1)-X(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))             354
152  001B0CI          P3=(X(I,J1,K1)-X(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))-             355     137
153                  1   (Y(I,J1,K1)-Y(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))             356
154  001E48I          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                     357     138
155  001E92I          P1=P1/PQ                                                       358     139
156  001EA4I          P2=P2/PQ                                                       359     140
157  001EB6I          P3=P3/PQ                                                       360     141
158  001EC8I          R1=(1.-P1**2)                                                  361     142
159  001EE4I          R2=(1.-P2**2)                                                  362     143
160  001F00I          R3=(1.-P3**2)                                                  363     144
161  001F1CI          SINX(III)=SQRT(R1)                                             364     145
162  001F42I          SINY(III)=SQRT(R2)                                             365     146
163  001F66I          SINZ(III)=SQRT(R3)                                             366     147
164  001F8AI          Q1=X(I,J,K)-X(I1,J,K)                                          367     148
165  001FDCI          Q2=Y(I,J,K)-Y(I1,J,K)                                          368     149
166  00202EI          Q3=Z(I,J,K)-Z(I1,J,K)                                          369     150
167  002080I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      370     151
168  0020FEI          CC=1.0                                                         371     152
169  00210AI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     372     153
170  002156I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             373     154
171  002198I          YN(III)=BB*ABS(COTH)                                           374     155
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172  0021C4I          Q1=X(I,J,K)-X(I2,J,K)                                    375     156
173  0022I6I          Q2=Y(I,J,K)-Y(I2,J,K)                                    376     157
174  0022C8I          Q3=Z(I,J,K)-Z(I2,J,K)                                    377     158
175  0022D8I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                               378     159
176  0023O6I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                379     160
177  0023B2I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                       380     161
178  0023C4I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                      381     162
179  0024O4I          IJLC(I,J,K)=III                                          382     163
180  0024O6I          III=III+1                                                383     164
181  0024BEI      4   CONTINUE                                                 384     165
182  0024BEI          IF(MC(I-1,J,K) .EQ. 0) GO TO 5                           385     166
183              C-----WEST                                                    386
184  0024C8I          I5C(III)=I                                               387     168
185  0024CCI          J8C(III)=J                                               388     169
186  0024D0I          K8C(III)=K                                               389     170
187  0024D4I          IITY(III)=4                                              390     171
188  0024D6I          J1=J+1                                                   391     172
189  0024D2I          J2=J-1                                                   392     173
190  0024E0I          K1=K+1                                                   393     174
191  0024E8I          K2=K-1                                                   394     175
192  0024FCI          IF(J .EQ. M1) J2=J                                       395     176
193  0025IAI          IF(J .EQ. M2) J1=J                                       396     178
194  0025I8I          IF(K .EQ. N1) K2=K                                       397     180
195  0025I6I          IF(K .EQ. N2) K1=K                                       398     182
196  0025I4I          I1=I+1                                                   399     184
197  0025I2I          I2=I+2                                                   400     185
198  0025I0I          P1=(Y(I,J1,K1)-Y(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))-       401     186
199                  1   (Z(I,J1,K1)-Z(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))       402
200  0026CCI          P2=(Z(I,J1,K1)-Z(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))-       403     187
201                  1   (X(I,J1,K1)-X(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))       404
202  0028O8I          P3=(X(I,J1,K1)-X(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))-       405     188
203                  1   (Y(I,J1,K1)-Y(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))       406
204  0029441          PQ=SQRT(P1*P1+P2*P2+P3*P3)                               407     189
205  00298EI          P1=P1/PQ                                                 408     190
206  0029A0I          P2=P2/PQ                                                 409     191
207  0029B2I          P3=P3/PQ                                                 410     192
208  0029C4I          R1=1.-P1**2                                              411     193
209  0029E0I          R2=1.-P2**2                                              412     194
210  0029FCI          R3=1.-P3**2                                              413     195
211  002A18I          SINX(III)=SQRT(R1)                                       414     196
212  002A3EI          SINY(III)=SQRT(R2)                                       415     197
213  002A62I          SINZ(III)=SQRT(R3)                                       416     198
214  002A86I          Q1=X(I,J,K)-X(I1,J,K)                                    417     199
215  002AD8I          Q2=Y(I,J,K)-Y(I1,J,K)                                    418     200
216  002B2AI          Q3=Z(I,J,K)-Z(I1,J,K)                                    419     201
217  002B7CI          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                420     202
218  002BFAI          CC=1.0                                                   421     203
219  002C06I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                               422     204
220  002C52I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                       423     205
221  002C94I          YN(III)=BB*ABS(COTH)                                     424     206
222  002CC0I          Q1=X(I,J,K)-X(I2,J,K)                                    425     207
223  002D12I          Q2=Y(I,J,K)-Y(I2,J,K)                                    426     208
224  002D64I          Q3=Z(I,J,K)-Z(I2,J,K)                                    427     209
225  002D86I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                               428     210
226  002E02I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                429     211
227  002E7EI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                       430     212
228  002EC0I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                      431     213
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229  002F00I          IJLC(I,J,K)=III                                          432  214
230  002F2CI          III=III+1                                                433  215
231  002F3AI        5 CONTINUE                                                 434  216
232  002F3AI          IF(MC(I,J,K+1) .EQ. 0) GO TO 6                           435  217
233                 C-----TOP                                                  436
234  002F72I          ISC(III)=I                                               437  219
235  002F86I          JBC(III)=J                                               438  220
236  002F9AI          KBC(III)=K                                               439  221
237  002FAEI          IITY(III)=5                                              440  222
238  002FBEI          I1=I+1                                                   441  223
239  002FCCI          I2=I-1                                                   442  224
240  002FDAI          J1=J+1                                                   443  225
241  002FE8I          J2=J-1                                                   444  226
242  002FF6I          IF(I .EQ. L1) I2=I                                       445  227
243  003014I          IF(I .EQ. L2) I1=I                                       446  229
244  003032I          IF(J .EQ. M1) J2=J                                       447  231
245  003050I          IF(J .EQ. M2) J1=J                                       448  233
246  00306EI          K1=K-1                                                   449  235
247  00307CI          K2=K-2                                                   450  236
248  00308AI          P1=(Y(I1,J1,K)-Y(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))-       451  237
249                  1   (Z(I1,J1,K)-Z(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))       452
250  0031C6I          P2=(Z(I1,J1,K)-Z(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))-       453  238
251                  1   (X(I1,J1,K)-X(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))       454
252  003302I          P3=(X(I1,J1,K)-X(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))-       455  239
253                  1   (Y(I1,J1,K)-Y(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))       456
254  00343EI          PQ=SQRT(P1*P1+P2*P2+P3*P3)                               457  240
255  00348AI          P1=P1/PQ                                                 458  241
256  00349CI          P2=P2/PQ                                                 459  242
257  0034AEI          P3=P3/PQ                                                 460  243
258  0034C0I          R1=1.-P1**2                                              461  244
259  0034DCI          R2=1.-P2**2                                              462  245
260  0034F8I          R3=1.-P3**2                                              463  246
261  003514I          SINX(III)=SQRT(R1)                                       464  247
262  00353AI          SINY(III)=SQRT(R2)                                       465  248
263  00355EI          SINZ(III)=SQRT(R3)                                       466  249
264  003582I          Q1=X(I,J,K)-X(I,J,K1)                                    467  250
265  0035D4I          Q2=Y(I,J,K)-Y(I,J,K1)                                    468  251
266  003626I          Q3=Z(I,J,K)-Z(I,J,K1)                                    469  252
267  003678I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                470  253
268  0036F6I          CC=1.0                                                   471  254
269  003702I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                               472  255
270  00374EI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                       473  256
271  003790I          YN(III)=BB*ABS(COTH)                                     474  257
272  0037BCI          Q1=X(I,J,K)-X(I,J,K2)                                    475  258
273  0037E0I          Q2=Y(I,J,K)-Y(I,J,K2)                                    476  259
274  003860I          Q3=Z(I,J,K)-Z(I,J,K2)                                    477  260
275  003882I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                               478  261
276  0038FEI          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                479  262
277  00397AI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                       480  263
278  0039BCI          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                      481  264
279  0039FCI          IJLC(I,J,K)=III                                          482  265
280  003A28I          III=III+1                                                483  266
281  003A36I        6 CONTINUE                                                 484  267
282  003A36I          IF(MC(I,J,K-1) .EQ. 0) GO TO 30                          485  268
283                 C-----BOTTOM                                               486
284  003A6EI          IBC(III)=I                                               487  270
285  003A82I          JBC(III)=J                                               488  271
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286  003A96I          KBC(III)=K                                                    489      272
287  003AAAI          IITY(III)=6                                                   490      273
288  003ABAI          I1=I+1                                                        491      274
289  003AC8I          I2=I-1                                                        492      275
290  003AD6I          J1=J+1                                                        493      276
291  003AE4I          J2=J-1                                                        494      277
292  003AF2I          IF(I .EQ. L1) I2=I                                            495      278
293  003B10I          IF(I .EQ. L2) I1=I                                            496      280
294  003B2EI          IF(J .EQ. M1) J2=J                                            497      282
295  003B4CI          IF(J .EQ. M2) J1=J                                            498      284
296  003B6AI          K1=K+1                                                        499      286
297  003B78I          K2=K+2                                                        500      287
298  003B86I          P1=(Y(I1,J1,K)-Y(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))-            501      288
299                  1    (Z(I1,J1,K)-Z(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))           502
300  003CC2I          P2=(Z(I1,J1,K)-Z(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))-            503      289
301                  1    (X(I1,J1,K)-X(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))           504
302  003DFEI          P3=(X(I1,J1,K)-X(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))-            505      290
303                  1    (Y(I1,J1,K)-Y(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))           506
304  003F3AI          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                    507      291
305  003F86I          P1=P1/PQ                                                      508      292
306  003F98I          P2=P2/PQ                                                      509      293
307  003FAAI          P3=P3/PQ                                                      510      294
308  003FBCI          R1=1.-P1**2                                                   511      295
309  003FD8I          R2=1.-P2**2                                                   512      296
310  003FF4I          R3=1.-P3**2                                                   513      297
311  004010I          SINX(III)=SQRT(R1)                                            514      298
312  004036I          SINY(III)=SQRT(R2)                                            515      299
313  00405AI          SINZ(III)=SQRT(R3)                                            516      300
314  00407EI          Q1=X(I,J,K)-X(I,J,K1)                                         517      301
315  004000I          Q2=Y(I,J,K)-Y(I,J,K1)                                         518      302
316  004122I          Q3=Z(I,J,K)-Z(I,J,K1)                                         519      303
317  004174I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                     520      304
318  0041F2I          CC=1.0                                                        521      305
319  0041FEI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                    522      306
320  00424AI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                            523      307
321  00428CI          YN(III)=BB*ABS(COTH)                                          524      308
322  004288I          Q1=X(I,J,K)-X(I,J,K2)                                         525      309
323  00430AI          Q2=Y(I,J,K)-Y(I,J,K2)                                         526      310
324  00435CI          Q3=Z(I,J,K)-Z(I,J,K2)                                         527      311
325  0043AEI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                    528      312
326  0043FAI          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                     529      313
327  004476I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                            530      314
328  004498I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                           531      315
329  0044F8I          IJLC(I,J,K)=III                                               532      316
330  004524I          III=III+1                                                     533      317
331  004532I     30   CONTINUE                                                      534      318
332  004580I          IITC=III-1                                                    535      319
333  00458EI          WRITE(6,100) L0,M0,N0,IITO                                    536      320
334  004598I    100   FORMAT(4I5)                                                   537
335  0045C4I          RETURN                                                        538      321
336  0045CCI          END                                                           539      322

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  CIRCCS    02/21/86  09:47:58  TABLE SPACE:   6 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 203 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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ORIGINAL PAGE IS
OF POOR QUALITY

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 1  000000I          SUBROUTINE TRANF                                                                  540
 2  0000004I         COMMON                                                                            541
 3                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),                           542
 4                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                                   543
 5                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                          544
 6                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                         545
 7                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                                 546
 8                  1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),               547
 9                  2 YN(800),YN1(800),SINX(800),SINY(300),SINZ(300),ANW1(800),                    548
10                  3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),                    549
11                  4 TALW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO                     550
12                  1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),                  551
13                  2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),                      552
14                  3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)                       553
15 0000004I          COMMON                                                                         554
16                  1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),                     555
17                  2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                                    556
18                  3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                                    557
19                  3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                                     558
20                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,                   559
21                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT                560
22 0000004I          COMMON                                                                         561
23                  1/TTRAN/TXXE(21,18,10),TXXW(21,18,10),TYYN(21,18,10),                          562
24                  2   TYYS(21,18,10),TZZT(21,18,10),TZZB(21,18,10),                              563
25                  3   TYXE(21,18,10),TYXW(21,18,10),TYZT(21,18,10),                              564
26                  4   TYZB(21,18,10),TXYN(21,18,10),TXYS(21,18,10),                              565
27                  5   TXZT(21,18,10),TXZB(21,18,10),TZXE(21,18,10),                              566
28                  6   TZXW(21,18,10),TZYN(21,18,10),TZYS(21,18,10)                               567
29                C-----CALCULATE GRID TRANSFORMATION COEFFICIENTS                                   568
30 0000004I          DO 40 I=2,L                                                                    569      1
31 0000018I          DO 40 J=2,M                                                                    570      2
32 00002CI           DO 40 K=2,N                                                                    571      3
33 000040I           P1=(X(I,J,K)+X(I,J-1,K)+X(I,J,K-1)+X(I,J-1,K-1)-X(I-1,J,K)-                    572      4
34                  1  X(I-1,J-1,K)-X(I-1,J,K-1)-X(I-1,J-1,K-1))*0.25                              573
35 0001A4I           P2=(X(I,J,K)+X(I-1,J,K)+X(I,J,K-1)+X(I-1,J,K-1)-X(I,J-1,K)-                    574      5
36                  1  X(I-1,J-1,K)-X(I,J-1,K-1)-X(I-1,J-1,K-1))*0.25                              575
37 000308I           P3=(X(I,J,K)+X(I-1,J,K)+X(I,J-1,K)+X(I-1,J-1,K)-X(I,J,K-1)-                    576      6
38                  1  X(I-1,J,K-1)-X(I,J-1,K-1)-X(I-1,J-1,K-1))*0.25                              577
39 00046CI           Q1=(Y(I,J,K)+Y(I,J-1,K)+Y(I,J,K-1)+Y(I,J-1,K-1)-Y(I-1,J,K)-                    578      7
40                  1  Y(I-1,J-1,K)-Y(I-1,J,K-1)-Y(I-1,J-1,K-1))*0.25                              579
41 0005D0I           Q2=(Y(I,J,K)+Y(I-1,J,K)+Y(I,J,K-1)+Y(I-1,J,K-1)-Y(I,J-1,K)-                    580      8
42                  1  Y(I-1,J-1,K)-Y(I,J-1,K-1)-Y(I-1,J-1,K-1))*0.25                              581
43 000734I           Q3=(Y(I,J,K)+Y(I-1,J,K)+Y(I,J-1,K)+Y(I-1,J-1,K)-Y(I,J,K-1)-                    582      9
44                  1  Y(I-1,J,K-1)-Y(I,J-1,K-1)-Y(I-1,J-1,K-1))*0.25                              583
45 0008F8I           R1=(Z(I,J,K)+Z(I,J-1,K)+Z(I,J,K-1)+Z(I,J-1,K-1)-Z(I-1,J,K)-                    584     10
46                  1  Z(I-1,J-1,K)-Z(I-1,J,K-1)-Z(I-1,J-1,K-1))*0.25                              585
47 0009FCI           R2=(Z(I,J,K)+Z(I-1,J,K)+Z(I,J,K-1)+Z(I-1,J,K-1)-Z(I,J-1,K)-                    586     11
48                  1  Z(I-1,J-1,K)-Z(I,J-1,K-1)-Z(I-1,J-1,K-1))*0.25                              587
49 000B60I           R3=(Z(I,J,K)+Z(I-1,J,K)+Z(I,J-1,K)+Z(I-1,J-1,K)-Z(I,J,K-1)-                    588     12
50                  1  Z(I-1,J,K-1)-Z(I,J-1,K-1)-Z(I-1,J-1,K-1))*0.25                              589
51 000CC4I           PTR=1.0/(P1*(Q2*R3-Q3*R2)-P2*(Q1*R3-Q3*R1)+P3*(Q1*R2-Q2*R1))                   590     13
52 000D36I           SU(I,J,K)=ABS(1.0/PTR)                                                         591     14
53 000D86I           AE(I,J,K)=PTR*(Q2*R3-Q3*R2)                                                    592     15
54 000DCCI           AW(I,J,K)=-PTR*(P2*R3-P3*R2)                                                   593     16
55 000E1AI           AN(I,J,K)=PTR*(P2*Q3-P3*Q2)                                                    594     17
56 000E60I           AS(I,J,K)=-PTR*(Q1*R3-Q3*R1)                                                   595     18
57 000EAEI           AT(I,J,K)=PTR*(P1*R3-P3*R1)                                                    596     19
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 58  000EF4I          AB(I,J,K)=-PTR*(P1*Q3-P3*Q1)                                                    597      20
 59  000F42I          DU(I,J,K)=PTR*(Q1*R2-Q2*R1)                                                     598      21
 60  000F88I          DV(I,J,K)=-PTR*(P1*R2-P2*R1)                                                    599      22
 61  000FD6I          DW(I,J,K)=PTR*(P1*Q2-P2*Q1)                                                     600      23
 62  00101CI      40  CONTINUE                                                                        601      24
 63  001064I          CALL WALVAL(1.0,2,L,2,M,2,N,AE)                                                 602      25
 64  0010BCI          CALL WALVAL(1.0,2,L,2,M,2,N,AW)                                                 603      26
 65  001114I          CALL WALVAL(1.0,2,L,2,M,2,N,AN)                                                 604      27
 66  00116CI          CALL WALVAL(1.0,2,L,2,M,2,N,AS)                                                 605      28
 67  0011C4I          CALL WALVAL(1.0,2,L,2,M,2,N,AT)                                                 606      29
 68  00121CI          CALL WALVAL(1.0,2,L,2,M,2,N,AB)                                                 607      30
 69  001274I          CALL WALVAL(1.0,2,L,2,M,2,N,DU)                                                 608      31
 70  0012CCI          CALL WALVAL(1.0,2,L,2,M,2,N,DV)                                                 609      32
 71  001324I          CALL WALVAL(1.0,2,L,2,M,2,N,DW)                                                 610      33
 72  00137CI          CALL WALVAL(1.0,2,L,2,M,2,N,SU)                                                 611      34
 73  0013D4I          DO 80 I=1,L                                                                     612      35
 74  0013E8I          DO 80 J=1,M                                                                     613      36
 75  0013FCI          DO 80 K=1,N                                                                     614      37
 76  001410I          CX(I,J,K)=(AE(I,J,K)+AE(I,J+1,K)+AE(I,J,K+1)+AE(I,J+1,K+1)+                     615      38
 77                  1 AE(I+1,J,K)+AE(I+1,J+1,K)+AE(I+1,J,K+1)+AE(I+1,J+1,K+1))*0.125                616
 78  001594I          CY(I,J,K)=(AW(I,J,K)+AW(I,J+1,K)+AW(I,J,K+1)+AW(I,J+1,K+1)+                     617      39
 79                  1 AW(I+1,J,K)+AW(I+1,J+1,K)+AW(I+1,J,K+1)+AW(I+1,J+1,K+1))*0.125                618
 80  001718I          CZ(I,J,K)=(AN(I,J,K)+AN(I,J+1,K)+AN(I,J,K+1)+AN(I,J+1,K+1)+                     619      40
 81                  1 AN(I+1,J,K)+AN(I+1,J+1,K)+AN(I+1,J,K+1)+AN(I+1,J+1,K+1))*0.125                620
 82  00189CI          EX(I,J,K)=(AS(I,J,K)+AS(I,J+1,K)+AS(I,J,K+1)+AS(I,J+1,K+1)+                     621      41
 83                  1 AS(I+1,J,K)+AS(I+1,J+1,K)+AS(I+1,J,K+1)+AS(I+1,J+1,K+1))*0.125                622
 84  001A20I          EY(I,J,K)=(AT(I,J,K)+AT(I,J+1,K)+AT(I,J,K+1)+AT(I,J+1,K+1)+                     623      42
 85                  1 AT(I+1,J,K)+AT(I+1,J+1,K)+AT(I+1,J,K+1)+AT(I+1,J+1,K+1))*0.125                624
 86  001BA4I          EZ(I,J,K)=(AB(I,J,K)+AB(I,J+1,K)+AB(I,J,K+1)+AB(I,J+1,K+1)+                     625      43
 87                  1 AB(I+1,J,K)+AB(I+1,J+1,K)+AB(I+1,J,K+1)+AB(I+1,J+1,K+1))*0.125                626
 88  001D28I          SX(I,J,K)=(DU(I,J,K)+DU(I,J+1,K)+DU(I,J,K+1)+DU(I,J+1,K+1)+                     627      44
 89                  1 DU(I+1,J,K)+DU(I+1,J+1,K)+DU(I+1,J,K+1)+DU(I+1,J+1,K+1))*0.125                628
 90  001EACI          SY(I,J,K)=(DV(I,J,K)+DV(I,J+1,K)+DV(I,J,K+1)+DV(I,J+1,K+1)+                     629      45
 91                  1 DV(I+1,J,K)+DV(I+1,J+1,K)+DV(I+1,J,K+1)+DV(I+1,J+1,K+1))*0.125                630
 92  002030I          SZ(I,J,K)=(DW(I,J,K)+DW(I,J+1,K)+DW(I,J,K+1)+DW(I,J+1,K+1)+                     631      46
 93                  1 DW(I+1,J,K)+DW(I+1,J+1,K)+DW(I+1,J,K+1)+DW(I+1,J+1,K+1))*0.125                632
 94  0021B4I          TJO(I,J,K)=(SU(I,J,K)+SU(I,J+1,K)+SU(I,J,K+1)+SU(I,J+1,K+1)+                    633      47
 95                  1 SU(I+1,J,K)+SU(I+1,J+1,K)+SU(I+1,J,K+1)+SU(I+1,J+1,K+1))*0.125                634
 96  002338I      80  CONTINUE                                                                        635      48
 97  002380I          DO 200 I=1,LO                                                                   636      49
 98  0023B4I          DO 200 J=1,MO                                                                   637      50
 99  0023A8I          DO 200 K=1,NO                                                                   638      51
100  0023BCI          DU(I,J,K)=0.0                                                                   639      52
101  0023E8I     200  CONTINUE                                                                        640      53
102  002430I          DO 160 I=2,LT                                                                   641      54
103  002444I          DO 160 J=2,MT                                                                   642      55
104  002458I          DO 160 K=2,NT                                                                   643      56
105  00246CI          CXE=(CX(I+1,J,K)+CX(I,J,K))*0.5                                                 644      57
106  0024C8I          CXW=(CX(I-1,J,K)+CX(I,J,K))*0.5                                                 645      58
107  002524I          CXN=(CX(I,J+1,K)+CX(I,J,K))*0.5                                                 646      59
108  002580I          CXS=(CX(I,J,K)+CX(I,J-1,K))*0.5                                                 647      60
109  0025DCI          CXT=(CX(I,J,K+1)+CX(I,J,K))*0.5                                                 648      61
110  002636I          CXB=(CX(I,J,K)+CX(I,J,K-1))*0.5                                                 649      62
111  002690I          CYE=(CY(I+1,J,K)+CY(I,J,K))*0.5                                                 650      63
112  0026ECI          CYW=(CY(I,J,K)+CY(I-1,J,K))*0.5                                                 651      64
113  002748I          CYN=(CY(I,J+1,K)+CY(I,J,K))*0.5                                                 652      65
114  0027A4I          CYS=(CY(I,J,K)+CY(I,J-1,K))*0.5                                                 653      66
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115  002800I        CYT=(CY(I,J,K+1)+CY(I,J,K))*0.5                                                    654   67
116  00285AI        CYB=(CY(I,J,K)+CY(I,J,K-1))*0.5                                                    655   68
117  0028B4I        CZE=(CZ(I+1,J,K)+CZ(I,J,K))*0.5                                                    656   69
118  002910I        CZW=(CZ(I,J,K)+CZ(I-1,J,K))*0.5                                                    657   70
119  00296CI        CZN=(CZ(I,J+1,K)+CZ(I,J,K))*0.5                                                    658   71
120  0029C8I        CZS=(CZ(I,J,K)+CZ(I,J-1,K))*0.5                                                    659   72
121  002A24I        CZT=(CZ(I,J,K+1)+CZ(I,J,K))*0.5                                                    660   73
122  002A7EI        CZB=(CZ(I,J,K)+CZ(I,J,K-1))*0.5                                                    661   74
123  002AD8I        EXE=(EX(I+1,J,K)+EX(I,J,K))*0.5                                                    662   75
124  002B34I        EXW=(EX(I,J,K)+EX(I-1,J,K))*0.5                                                    663   76
125  002B90I        EXN=(EX(I,J+1,K)+EX(I,J,K))*0.5                                                    664   77
126  002BECI        EXS=(EX(I,J,K)+EX(I,J-1,K))*0.5                                                    665   78
127  002C48I        EXT=(EX(I,J,K+1)+EX(I,J,K))*0.5                                                    666   79
128  002CA2I        EXB=(EX(I,J,K)+EX(I,J,K-1))*0.5                                                    667   80
129  002CFCI        EYE=(EY(I+1,J,K)+EY(I,J,K))*0.5                                                    668   81
130  002D58I        EYW=(EY(I,J,K)+EY(I-1,J,K))*0.5                                                    669   82
131  002DB4I        EYN=(EY(I,J+1,K)+EY(I,J,K))*0.5                                                    670   83
132  002E10I        EYS=(EY(I,J,K)+EY(I,J-1,K))*0.5                                                    671   84
133  002E6CI        EYT=(EY(I,J,K+1)+EY(I,J,K))*0.5                                                    672   85
134  002EC6I        EYB=(EY(I,J,K)+EY(I,J,K-1))*0.5                                                    673   86
135  002F20I        EZE=(EZ(I+1,J,K)+EZ(I,J,K))*0.5                                                    674   87
136  002F7CI        EZW=(EZ(I,J,K)+EZ(I-1,J,K))*0.5                                                    675   88
137  002FD8I        EZN=(EZ(I,J+1,K)+EZ(I,J,K))*0.5                                                    676   89
138  003034I        EZS=(EZ(I,J,K)+EZ(I,J-1,K))*0.5                                                    677   90
139  003090I        EZT=(EZ(I,J,K+1)+EZ(I,J,K))*0.5                                                    678   91
140  0030EAI        EZB=(EZ(I,J,K)+EZ(I,J,K-1))*0.5                                                    679   92
141  003144I        SXE=(SX(I+1,J,K)+SX(I,J,K))*0.5                                                    680   93
142  0031A0I        SXW=(SX(I,J,K)+SX(I-1,J,K))*0.5                                                    681   94
143  0031FCI        SXN=(SX(I,J+1,K)+SX(I,J,K))*0.5                                                    682   95
144  003258I        SXS=(SX(I,J,K)+SX(I,J-1,K))*0.5                                                    683   96
145  0032B4I        SXT=(SX(I,J,K+1)+SX(I,J,K))*0.5                                                    684   97
146  003310I        SXB=(SX(I,J,K)+SX(I,J,K-1))*0.5                                                    685   98
147  003368I        SYE=(SY(I+1,J,K)+SY(I,J,K))*0.5                                                    686   99
148  0033C4I        SYW=(SY(I,J,K)+SY(I-1,J,K))*0.5                                                    687  100
149  003420I        SYN=(SY(I,J+1,K)+SY(I,J,K))*0.5                                                    688  101
150  00347CI        SYS=(SY(I,J,K)+SY(I,J-1,K))*0.5                                                    689  102
151  0034D8I        SYT=(SY(I,J,K+1)+SY(I,J,K))*0.5                                                    690  103
152  003532I        SYB=(SY(I,J,K)+SY(I,J,K-1))*0.5                                                    691  104
153  00358CI        SZE=(SZ(I+1,J,K)+SZ(I,J,K))*0.5                                                    692  105
154  0035E8I        SZW=(SZ(I,J,K)+SZ(I-1,J,K))*0.5                                                    693  106
155  003644I        SZN=(SZ(I,J+1,K)+SZ(I,J,K))*0.5                                                    694  107
156  0036A0I        SZS=(SZ(I,J,K)+SZ(I,J-1,K))*0.5                                                    695  108
157  0036FCI        SZT=(SZ(I,J,K+1)+SZ(I,J,K))*0.5                                                    696  109
158  003756I        SZB=(SZ(I,J,K)+SZ(I,J,K-1))*0.5                                                    697  110
159  0037B0I        TXXE(I,J,K)=CXE*CX(I,J,K)+CYE*CY(I,J,K)+CZE*CZ(I,J,K)                              698  111
160  00385EI        TXXW(I,J,K)=CXW*CX(I,J,K)+CYW*CY(I,J,K)+CZW*CZ(I,J,K)                              699  112
161  00390CI        TYYN(I,J,K)=EXN*EX(I,J,K)+EYN*EY(I,J,K)+EZN*EZ(I,J,K)                              700  113
162  0039BAI        TYYS(I,J,K)=EXS*EX(I,J,K)+EYS*EY(I,J,K)+EZS*EZ(I,J,K)                              701  114
163  003A68I        TZZT(I,J,K)=SXT*SX(I,J,K)+SYT*SY(I,J,K)+SZT*SZ(I,J,K)                              702  115
164  003B16I        TZZB(I,J,K)=SXB*SX(I,J,K)+SYB*SY(I,J,K)+SZB*SZ(I,J,K)                              703  116
165  003BC4I        TYXE(I,J,K)=(EXE*CX(I,J,K)+EYE*CY(I,J,K)+EZE*CZ(I,J,K))*0.25                       704  117
166  003C78I        TYXW(I,J,K)=(EXW*CX(I,J,K)+EYW*CY(I,J,K)+EZW*CZ(I,J,K))*0.25                       705  118
167  003D2CI        TYZT(I,J,K)=(EXT*SX(I,J,K)+EYT*SY(I,J,K)+EZT*SZ(I,J,K))*0.25                       706  119
168  003DE0I        TYZB(I,J,K)=(EXB*SX(I,J,K)+EYB*SY(I,J,K)+EZB*SZ(I,J,K))*0.25                       707  120
169  003E94I        TXYN(I,J,K)=(CXN*EX(I,J,K)+CYN*EY(I,J,K)+CZN*EZ(I,J,K))*0.25                       708  121
170  003F48I        TXYS(I,J,K)=(CXS*EX(I,J,K)+CYS*EY(I,J,K)+CZS*EZ(I,J,K))*0.25                       709  122
171  003FFCI        TXZT(I,J,K)=(CXT*SX(I,J,K)+CYT*SY(I,J,K)+CZT*SZ(I,J,K))*0.25                       710  123
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172  0040B0I          TXZB(I,J,K)=(CXB*SX(I,J,K)+CYB*SY(I,J,K)+CZB*SZ(I,J,K))*0.25                711     124
173  0041A4I          TZXE(I,J,K)=(SXE*CX(I,J,K)+SYE*CY(I,J,K)+SZE*CZ(I,J,K))*0.25                712     125
174  0042I8I          TZXW(I,J,K)=(SXW*CX(I,J,K)+SYW*CY(I,J,K)+SZW*CZ(I,J,K))*0.25                713     126
175  0042CCI          TZYN(I,J,K)=(SXN*EX(I,J,K)+SYN*EY(I,J,K)+SZN*EZ(I,J,K))*0.25                714     127
176  0043B0I          TZYS(I,J,K)=(SXS*EX(I,J,K)+SYS*EY(I,J,K)+SZS*EZ(I,J,K))*0.25                715     128
177  0044A4I  160    CONTINUE                                                                     716     129
178  00447CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXXE)                                        717     130
179  0044D4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXXW)                                        718     131
180  00452CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYYN)                                        719     132
181  0045B4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYYS)                                        720     133
182  00465CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZZT)                                        721     134
183  0046E4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZZB)                                        722     135
184  00468CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYXE)                                        723     136
185  0046E4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYXW)                                        724     137
186  00473CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYZT)                                        725     138
187  0047B4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYZB)                                        726     139
188  0047ECI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXYN)                                        727     140
189  0048A4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXYS)                                        728     141
190  00489CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXZT)                                        729     142
191  0048F4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXZB)                                        730     143
192  00494CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZXE)                                        731     144
193  0049A4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZXW)                                        732     145
194  0049FCI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZYN)                                        733     146
195  004A54I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZYS)                                        734     147
196  004AACI          RETURN                                                                       735     148
197  004A84I          END                                                                          736     149

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  TRANF     02/21/86  09:50:44  TABLE SPACE:  10 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 199 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
  1  000000I          SUBROUTINE INIT                                                    737
  2  000004I          COMMON                                                             738
  3                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),            739
  4                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                    740
  5                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                          741
  6                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN             742
  7                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                     743
  8  000004I          COMMON                                                             744
  9                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LC,MC,ISWU,ISWV,ISWP,ISWK,ISWE,      745
 10                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NC,ISWW,IG,NT,ALC,DTT    746
 11             C-----INITIALIZE VARIABLES                                              747
 12  000004I          DO 10 I=1,LC                                                       748        1
 13  000018I          DO 10 J=1,MC                                                       749        2
 14  00002CI          DO 10 K=1,NC                                                       750        3
 15  000040I          U(I,J,K)=0.0                                                       751        4
 16  000054I          V(I,J,K)=0.0                                                       752        5
 17  000068I          W(I,J,K)=0.0                                                       753        6
 18  00007CI          P(I,J,K)=0.0                                                       754        7
 19  000090I          PP(I,J,K)=0.0                                                      755        8
 20  0000A4I          DK(I,J,K)=0.0                                                      756        9
 21  0000B8I          DE(I,J,K)=0.0                                                      757       10
 22  0000CCI          DEN(I,J,K)=DENIN                                                   758       11
 23  0000E0I          VISE(I,J,K)=VISC                                                   759       12
 24  0000F4I       10 CONTINUE                                                           760       13
 25  000108I          RETURN                                                             761       14
 26  00011AI          END                                                                762       15

NO  ERRORS:F7C  R05-01.0C  SUBROUTINE  INIT      02/21/86  09:51:07  TABLE SPACE:    3 KB
STATEMENT BUFFER:    20 LINES/1321 BYTES     STACK SPACE: 126 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
  1  000000I          SUBROUTINE NEWVIS                                                  763
  2  000004I          COMMON                                                             764
  3                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),            765
  4                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                    766
  5                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                          767
  6                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN             768
  7                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                     769
  8                  1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),   770
  9                  2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),        771
 10                  3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),        772
 11                  4 TALW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO         773
 12  000004I          COMMON                                                             774
 13                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,      775
 14                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT    776
 15                C-----EVALUATE TURBULENT VISCOSITY                                   777
 16  000004I          DO 10 I=1,L                                                        778        1
 17  000018I          DO 10 J=1,M                                                        779        2
 18  00002CI          DO 10 K=1,N                                                        780        3
 19  000040I          IF(DK(I,J,K) .LE. SMNUM) DK(I,J,K)=SMNUM                           781        4
 20  00009EI          IF(DE(I,J,K) .LE. SMNUM) DE(I,J,K)=SMNUM                           782        6
 21  0000FCI          IF(DE(I,J,K) .LE. SMNUM) GO TO 12      -                           783        8
 22  000134I          TURVIS=DEN(I,J,K)*CMU*DK(I,J,K)**2/DE(I,J,K)+VISC                  784       10
 23  0001C2I          GO TO 14                                                           785       11
 24  0001C8I     12   TURVIS=VISC                                                        786       12
 25  0001D4I     14   CONTINUE                                                           787       13
 26  0001D4I          VISE(I,J,K)=VISE(I,J,K)+ALVIS*(TURVIS-VISE(I,J,K))                 788       14
 27  000252I     10   CONTINUE                                                           789       15
 28  00029AI          RETURN                                                             790       16
 29  0002A0I          END                                                                791       17

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  NEWVIS    02/21/86  09:51:23  TABLE SPACE:   5 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 203 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
      SUBROUTINE SOLVEQ(IE,ISWF,ALF,SIGF,ERRF,F,F0)                                  792
      DIMENSION F(21,18,10),F1(21,18,10),F0(21,18,10)                                793
      COMMON                                                                         794
     1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),QK(21,18,10),                         795
     2  QE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                                  796
     3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                        797
     1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                         798
     1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                                 799
     1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),                800
     2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),                    801
     3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),                    802
     4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO                     803
     1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),                  804
     2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),                      805
     3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)                       806
      COMMON                                                                         807
     1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),                     808
     2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                                    809
     3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                                    810
     3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                                     811
     1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,                   812
     2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT                813
      COMMON                                                                         814
     1/TTRAN/TXXE(21,18,10),TXXW(21,18,10),TYYN(21,18,10),                          815
     2   TYYS(21,18,10),TZZT(21,18,10),TZZB(21,18,10),                              816
     3   TYXE(21,18,10),TYXW(21,18,10),TYZT(21,18,10),                              817
     4   TYZB(21,18,10),TXYN(21,18,10),TXYS(21,18,10),                              818
     5   TXZT(21,18,10),TXZB(21,18,10),TZXE(21,18,10),                              819
     6   TZXW(21,18,10),TZYN(21,18,10),TZYS(21,18,10)                               820
C------TRANSPORT EQUATIONS LINERAIZATION AND SOLVER                                 821
      ERRF=0.0                                                                       822        1
      PI=3.141592654                                                                 823        2
C-----PRESSURE CORRECTION SOLVER STARTS FROM 10                                      824
      IF(IE .EQ. 0) GO TO 10                                                         825        3
C-----U, V, W, TM, K & E EQUATIONS                                                  826
      IS=2                                                                           827        5
      IT=LT                                                                          828        6
      JS=2                                                                           829        7
      JT=MT                                                                          830        8
      KS=2                                                                           831        9
      KT=NT                                                                          832       10
      GO TO (21,29,29,29,29,21,21,21), IE                                            833       11
   21 CONTINUE                                                                       834       12
      DO 22 I=IS,IT                                                                  835       13
      DO 22 J=JS,JT                                                                  836       14
      DO 22 K=KS,KT                                                                  837       15
   22 F1(I,J,K)=VISE(I,J,K)/SIGF                                                     838       16
C-----EVALUATE LINK COEFF. AND SOURCE TERMS                                         839
      DO 20 I=IS,IT                                                                  840       17
      DO 20 J=JS,JT                                                                  841       18
      DO 20 K=KS,KT                                                                  842       19
      GAE=0.5*(F1(I+1,J,K)+F1(I,J,K))                                                843       20
      GAW=0.5*(F1(I-1,J,K)+F1(I,J,K))                                                844       21
      GAN=0.5*(F1(I,J+1,K)+F1(I,J,K))                                                845       22
      GAS=0.5*(F1(I,J-1,K)+F1(I,J,K))                                                846       23
      GAT=0.5*(F1(I,J,K+1)+F1(I,J,K))                                                847       24
      GAB=0.5*(F1(I,J,K-1)+F1(I,J,K))                                                848       25
```

```
 58  003F18I          DENE=0.5*(DEN(I+1,J,K)+DEN(I,J,K))                                  849      26
 59  003F74I          DENW=0.5*(DEN(I-1,J,K)+DEN(I,J,K))                                  850      27
 60  003FD0I          DENN=0.5*(DEN(I,J+1,K)+DEN(I,J,K))                                  851      28
 61  00402CI          DENS=0.5*(DEN(I,J-1,K)+DEN(I,J,K))                                  852      29
 62  004088I          DENT=0.5*(DEN(I,J,K+1)+DEN(I,J,K))                                  853      30
 63  0040E2I          DENB=0.5*(DEN(I,J,K-1)+DEN(I,J,K))                                  854      31
 64  00413CI          UE=C.5*(U(I+1,J,K)+U(I,J,K))                                        855      32
 65  004198I          UW=C.5*(U(I-1,J,K)+U(I,J,K))                                        856      33
 66  0041F4I          UN=C.5*(U(I,J+1,K)+U(I,J,K))                                        857      34
 67  004250I          US=C.5*(U(I,J-1,K)+U(I,J,K))                                        858      35
 68  0042ACI          UT=C.5*(U(I,J,K+1)+U(I,J,K))                                        859      36
 69  004306I          UB=0.5*(U(I,J,K-1)+U(I,J,K))                                        860      37
 70  004360I          VE=0.5*(V(I+1,J,K)+V(I,J,K))                                        861      38
 71  00433CI          VW=C.5*(V(I-1,J,K)+V(I,J,K))                                        862      39
 72  004418I          VN=C.5*(V(I,J+1,K)+V(I,J,K))                                        863      40
 73  004474I          VS=C.5*(V(I,J-1,K)+V(I,J,K))                                        864      41
 74  0044D0I          VT=C.5*(V(I,J,K+1)+V(I,J,K))                                        865      42
 75  00452AI          VB=C.5*(V(I,J,K-1)+V(I,J,K))                                        866      43
 76  004584I          WE=C.5*(W(I+1,J,K)+W(I,J,K))                                        867      44
 77  0045E0I          WW=C.5*(W(I-1,J,K)+W(I,J,K))                                        868      45
 78  00463CI          WN=C.5*(W(I,J+1,K)+W(I,J,K))                                        869      46
 79  004698I          WS=C.5*(W(I,J-1,K)+W(I,J,K))                                        870      47
 80  0046F4I          WT=C.5*(W(I,J,K+1)+W(I,J,K))                                        871      48
 81  00474EI          WB=C.5*(W(I,J,K-1)+W(I,J,K))                                        872      49
 82  0047A8I          CE=DENE*(UE*CX(I,J,K)+VE*CY(I,J,K)+WE*CZ(I,J,K))                    873      50
 83  00483CI          CW=DENW*(UW*CX(I,J,K)+VW*CY(I,J,K)+WW*CZ(I,J,K))                    874      51
 84  0048D0I          CN=DENN*(UN*EX(I,J,K)+VN*EY(I,J,K)+WN*EZ(I,J,K))                    875      52
 85  004964I          CS=DENS*(US*EX(I,J,K)+VS*EY(I,J,K)+WS*EZ(I,J,K))                    876      53
 86  0049F8I          CT=DENT*(UT*SX(I,J,K)+VT*SY(I,J,K)+WT*SZ(I,J,K))                    877      54
 87  004A8CI          CB=DENB*(UB*SX(I,J,K)+VB*SY(I,J,K)+WB*SZ(I,J,K))                    878      55
 88  004B20I          DDE=GAE*TXXE(I,J,K)+GAN*TXYN(I,J,K)-GAS*TXYS(I,J,K)+                879      56
 89                  1     GAT*TXZT(I,J,K)-GAB*TXZB(I,J,K)                               880
 90  004C0AI          DDW=GAW*TXXW(I,J,K)+GAS*TXYS(I,J,K)-GAN*TXYN(I,J,K)-                881      57
 91                  1     GAT*TXZT(I,J,K)+GAB*TXZB(I,J,K)                               882
 92  004CF4I          DDN=GAN*TYYN(I,J,K)+GAE*TYXE(I,J,K)-GAW*TYXW(I,J,K)+                883      58
 93                  1     GAT*TYZT(I,J,K)-GAB*TYZB(I,J,K)                               884
 94  004DDEI          DDS=GAS*TYYS(I,J,K)+GAW*TYXW(I,J,K)-GAE*TYXE(I,J,K)-                885      59
 95                  1     GAT*TYZT(I,J,K)+GAB*TYZB(I,J,K)                               886
 96  004EC8I          DDT=GAT*TZZT(I,J,K)+GAE*TZXE(I,J,K)-GAW*TZXW(I,J,K)+                887      60
 97                  1     GAN*TZYN(I,J,K)-GAS*TZYS(I,J,K)                               888
 98  004FB2I          DDB=GAB*TZZB(I,J,K)+GAW*TZXW(I,J,K)-GAE*TZXE(I,J,K)-                889      61
 99                  1     GAN*TZYN(I,J,K)+GAS*TZYS(I,J,K)                               890
100  00509CI          CPO=ABS(CE-CW+CN-CS+CT-CB)                                          891      62
101  0050E2I          AE(I,J,K)=(AMAX1(ABS(0.5*CE),DDE)-0.5*CE)*TJO(I,J,K)                892      63
102  005186I          AW(I,J,K)=(AMAX1(ABS(0.5*CW),DDW)+0.5*CW)*TJO(I,J,K)                893      64
103  00522AI          AN(I,J,K)=(AMAX1(ABS(0.5*CN),DDN)-0.5*CN)*TJO(I,J,K)                894      65
104  0052CEI          AS(I,J,K)=(AMAX1(ABS(0.5*CS),DDS)+0.5*CS)*TJO(I,J,K)                895      66
105  005372I          AT(I,J,K)=(AMAX1(ABS(0.5*CT),DDT)-0.5*CT)*TJO(I,J,K)                896      67
106  005416I          AB(I,J,K)=(AMAX1(ABS(0.5*CB),DDB)+0.5*CB)*TJO(I,J,K)                897      68
107  0054BAI          DDNE=GAE*TYXE(I,J,K)+GAN*TXYN(I,J,K)                                898      69
108  00551AI          DDSE=-GAE*TYXE(I,J,K)-GAS*TXYS(I,J,K)                               899      70
109  005582I          DDNW=-GAW*TYXW(I,J,K)-GAN*TXYN(I,J,K)                               900      71
110  0055EAI          DDSW=GAW*TYXW(I,J,K)+GAS*TXYS(I,J,K)                                901      72
111  00564AI          DDET=GAE*TZXE(I,J,K)+GAT*TXZT(I,J,K)                                902      73
112  0056AAI          DDEB=-GAE*TZXE(I,J,K)-GAB*TXZB(I,J,K)                               903      74
113  005712I          DDWT=-GAW*TZXW(I,J,K)-GAT*TXZT(I,J,K)                               904      75
114  00577AI          DDWB=GAW*TZXW(I,J,K)+GAB*TXZB(I,J,K)                                905      76
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ORIGINAL PAGE IS
OF POOR QUALITY

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      DONT=GAN*TZYN(I,J,K)+GAT*TYZT(I,J,K)                                  906      77
      DONB=-GAN*TZYN(I,J,K)-GAB*TYZB(I,J,K)                                 907      78
      DOST=-GAS*TZYS(I,J,K)-GAT*TYZT(I,J,K)                                 908      79
      DOSB=GAS*TZYS(I,J,K)+GAB*TYZB(I,J,K)                                  909      80
      SU(I,J,K)=CPO*F(I,J,K)+DONE*F(I+1,J+1,K)+DOSE*F(I+1,J-1,K)+           910      81
     1          DONW*F(I-1,J+1,K)+DOSW*F(I-1,J-1,K)+                        911
     2          DOET*F(I+1,J,K+1)+DOEB*F(I+1,J,K-1)+                        912
     3          DOWT*F(I-1,J,K+1)+DOWB*F(I-1,J,K-1)+                        913
     4          DONT*F(I,J+1,K+1)+DONB*F(I,J+1,K-1)+                        914
     5          DOST*F(I,J-1,K+1)+DOSB*F(I,J-1,K-1)                         915
      SV(I,J,K)=CPO*V(I,J,K)+DONE*V(I+1,J+1,K)+DOSE*V(I+1,J-1,K)+           916      82
     1          DONW*V(I-1,J+1,K)+DOSW*V(I-1,J-1,K)+                        917
     2          DOET*V(I+1,J,K+1)+DOEB*V(I+1,J,K-1)+                        918
     3          DOWT*V(I-1,J,K+1)+DOWB*V(I-1,J,K-1)+                        919
     4          DONT*V(I,J+1,K+1)+DONB*V(I,J+1,K-1)+                        920
     5          DOST*V(I,J-1,K+1)+DOSB*V(I,J-1,K-1)                         921
      SW(I,J,K)=CPO*W(I,J,K)+DONE*W(I+1,J+1,K)+DOSE*W(I+1,J-1,K)+           922      83
     1          DONW*W(I-1,J+1,K)+DOSW*W(I-1,J-1,K)+                        923
     2          DOET*W(I+1,J,K+1)+DOEB*W(I+1,J,K-1)+                        924
     3          DOWT*W(I-1,J,K+1)+DOWB*W(I-1,J,K-1)+                        925
     4          DONT*W(I,J+1,K+1)+DONB*W(I,J+1,K-1)+                        926
     5          DOST*W(I,J-1,K+1)+DOSB*W(I,J-1,K-1)                         927
      PP(I,J,K)=CPO*DK(I,J,K)+DONE*DK(I+1,J+1,K)+DOSE*DK(I+1,J-1,K)+        928      84
     1          DONW*DK(I-1,J+1,K)+DOSW*DK(I-1,J-1,K)+                      929
     2          DOET*DK(I+1,J,K+1)+DOEB*DK(I+1,J,K-1)+                      930
     3          DOWT*DK(I-1,J,K+1)+DOWB*DK(I-1,J,K-1)+                      931
     4          DONT*DK(I,J+1,K+1)+DONB*DK(I,J+1,K-1)+                      932
     5          DOST*DK(I,J-1,K+1)+DOSB*DK(I,J-1,K-1)                       933
      SUK(I,J,K)=CPO                                                        934      85
      SPK(I,J,K)=-CPO-APC(I,J,K)                                            935      86
      UCXI=UE-UW                                                            936      87
      UEDA=UN-US                                                            937      88
      USCI=UT-UB                                                            938      89
      VCXI=VE-VW                                                            939      90
      VEDA=VN-VS                                                            940      91
      VSCI=VT-VB                                                            941      92
      WCXI=WE-WW                                                            942      93
      WEDA=WN-WS                                                            943      94
      WSCI=WT-WB                                                            944      95
      UX=UCXI*CX(I,J,K)+UEDA*EX(I,J,K)+USCI*SX(I,J,K)                       945      96
      UY=UCXI*CY(I,J,K)+UEDA*EY(I,J,K)+USCI*SY(I,J,K)                       946      97
      UZ=UCXI*CZ(I,J,K)+UEDA*EZ(I,J,K)+USCI*SZ(I,J,K)                       947      98
      VX=VCXI*CX(I,J,K)+VEDA*EX(I,J,K)+VSCI*SX(I,J,K)                       948      99
      VY=VCXI*CY(I,J,K)+VEDA*EY(I,J,K)+VSCI*SY(I,J,K)                       949     100
      VZ=VCXI*CZ(I,J,K)+VEDA*EZ(I,J,K)+VSCI*SZ(I,J,K)                       950     101
      WX=WCXI*CX(I,J,K)+WEDA*EX(I,J,K)+WSCI*SX(I,J,K)                       951     102
      WY=WCXI*CY(I,J,K)+WEDA*EY(I,J,K)+WSCI*SY(I,J,K)                       952     103
      WZ=WCXI*CZ(I,J,K)+WEDA*EZ(I,J,K)+WSCI*SZ(I,J,K)                       953     104
      GEN(I,J,K)=VISE(I,J,K)*((UY+VX)**2+(VZ+WY)**2+(WX+UZ)**2+             954     105
     1          2*(UX*UX+VY*VY+WZ*WZ))                                      955
   20 CONTINUE                                                              956     106
C-------------------------------------------------------------------------  957
   29 CONTINUE                                                              958     107
C-----CALCULATE SOURCE TERMS                                                959
      GO TO (1,2,3,4,5,6,7), IE                                             960     108
C-----U-, V-, W-SOURCES                                                     961
    1 CONTINUE                                                              962     109
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```
172  006C66I          DO 15 I=IS,IT                                                  963     110
173  006C7EI          DO 15 J=JS,JT                                                  964     111
174  006C96I          DO 15 K=KS,KT                                                  965     112
175  006CAEI          GAE=0.5*(F1(I+1,J,K)+F1(I,J,K))                                966     113
176  006DOAI          GAW=0.5*(F1(I-1,J,K)+F1(I,J,K))                                967     114
177  006D66I          GAN=0.5*(F1(I,J+1,K)+F1(I,J,K))                                968     115
178  006DC2I          GAS=0.5*(F1(I,J-1,K)+F1(I,J,K))                                969     116
179  006E1EI          GAT=0.5*(F1(I,J,K+1)+F1(I,J,K))                                970     117
180  006E78I          GAB=0.5*(F1(I,J,K-1)+F1(I,J,K))                                971     118
181  006ED2I          PCXI=(P(I+1,J,K)+P(I+1,J+1,K)+P(I+1,J,K+1)+P(I+1,J+1,K+1)-     972     119
182                  1       P(I,J,K)-P(I,J+1,K)-P(I,J,K+1)-P(I,J+1,K+1))*0.25       973
183  0070361          PEDA=(P(I,J+1,K)+P(I+1,J+1,K)+P(I,J+1,K+1)+P(I+1,J+1,K+1)-     974     120
184                  1       P(I,J,K)-P(I+1,J,K)-P(I,J,K+1)-P(I+1,J,K+1))*0.25       975
185  00719AI          PSCI=(P(I,J,K+1)+P(I+1,J,K+1)+P(I,J+1,K+1)+P(I+1,J+1,K+1)-     976     121
186                  1       P(I,J,K)-P(I+1,J,K)-P(I,J+1,K)-P(I+1,J+1,K))*0.25       977
187  0072FEI          UCE=U(I+1,J,K)-U(I,J,K)                                        978     122
188  0073541          UCW=U(I,J,K)-U(I-1,J,K)                                        979     123
189  0073AAI          UCN=(U(I+1,J+1,K)+U(I+1,J,K)-U(I-1,J+1,K)-U(I-1,J,K))*0.25     980     124
190  0074661          UCS=(U(I+1,J,K)+U(I+1,J-1,K)-U(I-1,J,K)-U(I-1,J-1,K))*0.25     981     125
191  0075221          UCT=(U(I+1,J,K+1)+U(I+1,J,K)-U(I-1,J,K+1)-U(I-1,J,K))*0.25     982     126
192  0075DAI          UCB=(U(I+1,J,K)+U(I+1,J,K-1)-U(I-1,J,K)-U(I-1,J,K-1))*0.25     983     127
193  0076921          UEE=(U(I+1,J+1,K)+U(I,J+1,K)-U(I+1,J-1,K)-U(I,J-1,K))*0.25     984     128
194  00774EI          UEW=(U(I,J+1,K)+U(I-1,J+1,K)-U(I,J-1,K)-U(I-1,J-1,K))*0.25     985     129
195  00780AI          UEN=U(I,J+1,K)-U(I,J,K)                                        986     130
196  0078601          UES=U(I,J,K)-U(I,J-1,K)                                        987     131
197  0078861          UET=(U(I,J+1,K+1)+U(I,J+1,K)-U(I,J-1,K+1)-U(I,J-1,K))*0.25     988     132
198  00796EI          UEB=(U(I,J+1,K)+U(I,J+1,K-1)-U(I,J-1,K)-U(I,J-1,K-1))*0.25     989     133
199  007A261          USE=(U(I+1,J,K+1)+U(I,J,K+1)-U(I+1,J,K-1)-U(I,J,K-1))*0.25     990     134
200  007ADAI          USW=(U(I,J,K+1)+U(I-1,J,K+1)-U(I,J,K-1)-U(I-1,J,K-1))*0.25     991     135
201  007B3E1          USN=(U(I,J+1,K+1)+U(I,J,K+1)-U(I,J+1,K-1)-U(I,J,K-1))*0.25     992     136
202  007C421          USS=(U(I,J,K+1)+U(I,J-1,K+1)-U(I,J,K-1)-U(I,J-1,K-1))*0.25     993     137
203  007CF61          UST=U(I,J,K+1)-U(I,J,K)                                        994     138
204  007D4AI          USB=U(I,J,K)-U(I,J,K-1)                                        995     139
205  007D9EI          VCE=V(I+1,J,K)-V(I,J,K)                                        996     140
206  007DF41          VCW=V(I,J,K)-V(I-1,J,K)                                        997     141
207  007E4AI          VCN=(V(I+1,J+1,K)+V(I+1,J,K)-V(I-1,J+1,K)-V(I-1,J,K))*0.25     998     142
208  007F061          VCS=(V(I+1,J,K)+V(I+1,J-1,K)-V(I-1,J,K)-V(I-1,J-1,K))*0.25     999     143
209  007F9C1          VCT=(V(I+1,J,K+1)+V(I+1,J,K)-V(I-1,J,K+1)-V(I-1,J,K))*0.25     1000    144
210  00807AI          VCB=(V(I+1,J,K)+V(I+1,J,K-1)-V(I-1,J,K)-V(I-1,J,K-1))*0.25     1001    145
211  0081321          VEE=(V(I+1,J+1,K)+V(I,J+1,K)-V(I+1,J-1,K)-V(I,J-1,K))*0.25     1002    146
212  0081EEI          VEW=(V(I,J+1,K)+V(I-1,J+1,K)-V(I,J-1,K)-V(I-1,J-1,K))*0.25     1003    147
213  0082AAI          VEN=V(I,J+1,K)-V(I,J,K)                                        1004    148
214  0083001          VES=V(I,J,K)-V(I,J-1,K)                                        1005    149
215  0083561          VET=(V(I,J+1,K+1)+V(I,J+1,K)-V(I,J-1,K+1)-V(I,J-1,K))*0.25     1006    150
216  00840EI          VEB=(V(I,J+1,K)+V(I,J+1,K-1)-V(I,J-1,K)-V(I,J-1,K-1))*0.25     1007    151
217  0084C61          VSE=(V(I+1,J,K+1)+V(I,J,K+1)-V(I+1,J,K-1)-V(I,J,K-1))*0.25     1008    152
218  00857AI          VSW=(V(I,J,K+1)+V(I-1,J,K+1)-V(I,J,K-1)-V(I-1,J,K-1))*0.25     1009    153
219  0086261          VSN=(V(I,J+1,K+1)+V(I,J,K+1)-V(I,J+1,K-1)-V(I,J,K-1))*0.25     1010    154
220  0086E21          VSS=(V(I,J,K+1)+V(I,J-1,K+1)-V(I,J,K-1)-V(I,J-1,K-1))*0.25     1011    155
221  0087961          VST=V(I,J,K+1)-V(I,J,K)                                        1012    156
222  0087EAI          VSB=V(I,J,K)-V(I,J,K-1)                                        1013    157
223  00883EI          WCE=W(I+1,J,K)-W(I,J,K)                                        1014    158
224  0088941          WCW=W(I,J,K)-W(I-1,J,K)                                        1015    159
225  0088EAI          WCN=(W(I+1,J+1,K)+W(I+1,J,K)-W(I-1,J+1,K)-W(I-1,J,K))*0.25     1016    160
226  0089A61          WCS=(W(I+1,J,K)+W(I+1,J-1,K)-W(I-1,J,K)-W(I-1,J-1,K))*0.25     1017    161
227  008A621          WCT=(W(I+1,J,K+1)+W(I+1,J,K)-W(I-1,J,K+1)-W(I-1,J,K))*0.25     1018    162
228  008B1AI          WCB=(W(I+1,J,K)+W(I+1,J,K-1)-W(I-1,J,K)-W(I-1,J,K-1))*0.25     1019    163
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229  0086D2I        WEE=(W(I+1,J+1,K)+W(I,J+1,K)-W(I+1,J-1,K)-W(I,J-1,K))*0.25                                1020    164
230  0086E8I        WEW=(W(I,J+1,K)+W(I-1,J+1,K)-W(I,J-1,K)-W(I-1,J-1,K))*0.25                               1021    165
231  0086D4AI       WEN=W(I,J+1,K)-W(I,J,K)                                                                  1022    166
232  0086DA0I       WES=W(I,J,K)-W(I,J-1,K)                                                                  1023    167
233  0086DF6I       WET=(W(I,J+1,K+1)+W(I,J+1,K)-W(I,J-1,K+1)-W(I,J-1,K))*0.25                               1024    168
234  0086EAEI       WEB=(W(I,J+1,K)+W(I,J+1,K-1)-W(I,J-1,K)-W(I,J-1,K-1))*0.25                               1025    169
235  0086F66I       WSE=(W(I+1,J,K+1)+W(I,J,K+1)-W(I+1,J,K-1)-W(I,J,K-1))*0.25                               1026    170
236  00901AI        WSW=(W(I,J,K+1)+W(I-1,J,K+1)-W(I,J,K-1)-W(I-1,J,K-1))*0.25                               1027    171
237  0090CEI        WSN=(W(I,J+1,K+1)+W(I,J,K+1)-W(I,J+1,K-1)-W(I,J,K-1))*0.25                               1028    172
238  009182I        WSS=(W(I,J,K+1)+W(I,J-1,K+1)-W(I,J,K-1)-W(I,J-1,K-1))*0.25                               1029    173
239  009236I        WST=W(I,J,K+1)-W(I,J,K)                                                                  1030    174
240  00928AI        WSB=W(I,J,K)-W(I,J,K-1)                                                                  1031    175
241  0092DEI        SU(I,J,K)=SU(I,J,K)-PCXI*CX(I,J,K)-PEDA*EX(I,J,K)-PSCI*SX(I,J,K)                         1032    176
242  009384I        DV(I,J,K)=DV(I,J,K)-PCXI*CY(I,J,K)-PEDA*EY(I,J,K)-PSCI*SY(I,J,K)                         1033    177
243  00948AI        DW(I,J,K)=DW(I,J,K)-PCXI*CZ(I,J,K)-PEDA*EZ(I,J,K)-PSCI*SZ(I,J,K)                         1034    178
244  009560I        CXE=(CX(I+1,J,K)+CX(I,J,K))*0.5                                                          1035    179
245  009596I        CXW=(CX(I-1,J,K)+CX(I,J,K))*0.5                                                          1036    180
246  0096I8I        CXN=(CX(I,J+1,K)+CX(I,J,K))*0.5                                                          1037    181
247  009674I        CXS=(CX(I,J,K)+CX(I,J-1,K))*0.5                                                          1038    182
248  0096D0I        CXT=(CX(I,J,K+1)+CX(I,J,K))*0.5                                                          1039    183
249  009726I        CXB=(CX(I,J,K)+CX(I,J,K-1))*0.5                                                          1040    184
250  009784I        EXE=(EX(I+1,J,K)+EX(I,J,K))*0.5                                                          1041    185
251  0097E0I        EXW=(EX(I,J,K)+EX(I-1,J,K))*0.5                                                          1042    186
252  009836I        EXN=(EX(I,J+1,K)+EX(I,J,K))*0.5                                                          1043    187
253  009898I        EXS=(EX(I,J,K)+EX(I,J-1,K))*0.5                                                          1044    188
254  0098F4I        EXT=(EX(I,J,K+1)+EX(I,J,K))*0.5                                                          1045    189
255  00994EI        EXB=(EX(I,J,K)+EX(I,J,K-1))*0.5                                                          1046    190
256  0099A8I        SXE=(SX(I+1,J,K)+SX(I,J,K))*0.5                                                          1047    191
257  009A04I        SXW=(SX(I,J,K)+SX(I-1,J,K))*0.5                                                          1048    192
258  009A60I        SXN=(SX(I,J+1,K)+SX(I,J,K))*0.5                                                          1049    193
259  009ABCI        SXS=(SX(I,J,K)+SX(I,J-1,K))*0.5                                                          1050    194
260  009B18I        SXT=(SX(I,J,K+1)+SX(I,J,K))*0.5                                                          1051    195
261  009B72I        SXB=(SX(I,J,K)+SX(I,J,K-1))*0.5                                                          1052    196
262  009BCCI        QE=GAE*(UCE*CXE+UEE*EXE+USE*SXE)                                                         1053    197
263  009C00I        QW=GAW*(UCW*CXW+UEW*EXW+USW*SXW)                                                         1054    198
264  009C34I        QN=GAN*(UCN*CXN+UEN*EXN+USN*SXN)                                                         1055    199
265  009C68I        QS=GAS*(UCS*CXS+UES*EXS+USS*SXS)                                                         1056    200
266  009C9CI        QT=GAT*(UCT*CXT+UET*EXT+UST*SXT)                                                         1057    201
267  009CD0I        QB=GAB*(UCB*CXB+UEB*EXB+USB*SXB)                                                         1058    202
268  009D04I        SOC1=CX(I,J,K)*(QE-QW)+EX(I,J,K)*(QN-QS)+SX(I,J,K)*(QT-QB)                               1059    203
269  009DA4I        QE=GAE*(VCE*CXE+VEE*EXE+VSE*SXE)                                                         1060    204
270  009DD8I        QW=GAW*(VCW*CXW+VEW*EXW+VSW*SXW)                                                         1061    205
271  009E0CI        QN=GAN*(VCN*CXN+VEN*EXN+VSN*SXN)                                                         1062    206
272  009E40I        QS=GAS*(VCS*CXS+VES*EXS+VSS*SXS)                                                         1063    207
273  009E74I        QT=GAT*(VCT*CXT+VET*EXT+VST*SXT)                                                         1064    208
274  009EA8I        QB=GAB*(VCB*CXB+VEB*EXB+VSB*SXB)                                                         1065    209
275  009EDCI        SOC2=CY(I,J,K)*(QE-QW)+EY(I,J,K)*(QN-QS)+SY(I,J,K)*(QT-QB)                               1066    210
276  009F7CI        QE=GAE*(WCE*CXE+WEE*EXE+WSE*SXE)                                                         1067    211
277  009FB0I        QW=GAW*(WCW*CXW+WEW*EXW+WSW*SXW)                                                         1068    212
278  009FE4I        QN=GAN*(WCN*CXN+WEN*EXN+WSN*SXN)                                                         1069    213
279  00A018I        QS=GAS*(WCS*CXS+WES*EXS+WSS*SXS)                                                         1070    214
280  00A04CI        QT=GAT*(WCT*CXT+WET*EXT+WST*SXT)                                                         1071    215
281  00A080I        QB=GAB*(WCB*CXB+WEB*EXB+WSB*SXB)                                                         1072    216
282  00A0B4I        SOC3=CZ(I,J,K)*(QE-QW)+EZ(I,J,K)*(QN-QS)+SZ(I,J,K)*(QT-QB)                               1073    217
283  00A154I        SU(I,J,K)=SU(I,J,K)+SOC1+SOC2+SOC3+APO(I,J,K)*FO(I,J,K)                                  1074    218
284  00A206I        CYE=(CY(I+1,J,K)+CY(I,J,K))*0.5                                                          1075    219
285  00A262I        CYW=(CY(I,J,K)+CY(I-1,J,K))*0.5                                                          1076    220
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286  00A28EI          CYN=(CY(I,J+1,K)+CY(I,J,K))*0.5                                                    1077     221
287  00A31AI          CYS=(CY(I,J,K)+CY(I,J-1,K))*0.5                                                    1078     222
288  00A376I          CYT=(CY(I,J,K+1)+CY(I,J,K))*0.5                                                    1079     223
289  00A300I          CYB=(CY(I,J,K)+CY(I,J,K-1))*0.5                                                    1080     224
290  00A42AI          EYE=(EY(I+1,J,K)+EY(I,J,K))*0.5                                                    1081     225
291  00A486I          EYW=(EY(I,J,K)+EY(I-1,J,K))*0.5                                                    1082     226
292  00A4E2I          EYN=(EY(I,J+1,K)+EY(I,J,K))*0.5                                                    1083     227
293  00A53EI          EYS=(EY(I,J,K)+EY(I,J-1,K))*0.5                                                    1084     228
294  00A59AI          EYT=(EY(I,J,K+1)+EY(I,J,K))*0.5                                                    1085     229
295  00A5F4I          EYB=(EY(I,J,K)+EY(I,J,K-1))*0.5                                                    1086     230
296  00A64EI          SYE=(SY(I+1,J,K)+SY(I,J,K))*0.5                                                    1087     231
297  00A6AAI          SYW=(SY(I,J,K)+SY(I-1,J,K))*0.5                                                    1088     232
298  00A706I          SYN=(SY(I,J+1,K)+SY(I,J,K))*0.5                                                    1089     233
299  00A762I          SYS=(SY(I,J,K)+SY(I,J-1,K))*0.5                                                    1090     234
300  00A7BEI          SYT=(SY(I,J,K+1)+SY(I,J,K))*0.5                                                    1091     235
301  00A818I          SYB=(SY(I,J,K)+SY(I,J,K-1))*0.5                                                    1092     236
302  00A872I          QE=GAE*(UCE*CYE+UEE*EYE+USE*SYE)                                                   1093     237
303  00A8A6I          QW=GAW*(UCW*CYW+UEW*EYW+USW*SYW)                                                   1094     238
304  00A8DAI          QN=GAN*(UCN*CYN+UEN*EYN+USN*SYN)                                                   1095     239
305  00A90EI          QS=GAS*(UCS*CYS+UES*EYS+USS*SYS)                                                   1096     240
306  00A942I          QT=GAT*(UCT*CYT+UET*EYT+UST*SYT)                                                   1097     241
307  00A976I          QB=GAB*(UCB*CYB+UEB*EYB+USB*SYB)                                                   1098     242
308  00A9AAI          SOC1=CX(I,J,K)*(QE-QW)+EX(I,J,K)*(QN-QS)+SX(I,J,K)*(QT-QB)                         1099     243
309  00AA4AI          QE=GAE*(VCE*CYE+VEE*EYE+VSE*SYE)                                                   1100     244
310  00AA7EI          QW=GAW*(VCW*CYW+VEW*EYW+VSW*SYW)                                                   1101     245
311  00AAB2I          QN=GAN*(VCN*CYN+VEN*EYN+VSN*SYN)                                                   1102     246
312  00AAE6I          QS=GAS*(VCS*CYS+VES*EYS+VSS*SYS)                                                   1103     247
313  00AB1AI          QT=GAT*(VCT*CYT+VET*EYT+VST*SYT)                                                   1104     248
314  00AB4EI          QB=GAB*(VCB*CYB+VEB*EYB+VSB*SYB)                                                   1105     249
315  00AB82I          SOC2=CY(I,J,K)*(QE-QW)+EY(I,J,K)*(QN-QS)+SY(I,J,K)*(QT-QB)                         1106     250
316  00AC22I          QE=GAE*(WCE*CYE+WEE*EYE+WSE*SYE)                                                   1107     251
317  00AC56I          QW=GAW*(WCW*CYW+WEW*EYW+WSW*SYW)                                                   1108     252
318  00AC8AI          QN=GAN*(WCN*CYN+WEN*EYN+WSN*SYN)                                                   1109     253
319  00ACBEI          QS=GAS*(WCS*CYS+WES*EYS+WSS*SYS)                                                   1110     254
320  00ACF2I          QT=GAT*(WCT*CYT+WET*EYT+WST*SYT)                                                   1111     255
321  00AD26I          QB=GAB*(WCB*CYB+WEB*EYB+WSB*SYB)                                                   1112     256
322  00AD5AI          SOC3=CZ(I,J,K)*(QE-QW)+EZ(I,J,K)*(QN-QS)+SZ(I,J,K)*(QT-QB)                         1113     257
323  00AD9AI          DV(I,J,K)=DV(I,J,K)+SOC1+SOC2+SOC3                                                 1114     258
324  00AE58I          CZE=(CZ(I+1,J,K)+CZ(I,J,K))*0.5                                                    1115     259
325  00AE94I          CZW=(CZ(I,J,K)+CZ(I-1,J,K))*0.5                                                    1116     260
326  00AF10I          CZN=(CZ(I,J+1,K)+CZ(I,J,K))*0.5                                                    1117     261
327  00AF6CI          CZS=(CZ(I,J,K)+CZ(I,J-1,K))*0.5                                                    1118     262
328  00AFC8I          CZT=(CZ(I,J,K+1)+CZ(I,J,K))*0.5                                                    1119     263
329  00B022I          CZB=(CZ(I,J,K)+CZ(I,J,K-1))*0.5                                                    1120     264
330  00B07CI          EZE=(EZ(I+1,J,K)+EZ(I,J,K))*0.5                                                    1121     265
331  00B0D8I          EZW=(EZ(I,J,K)+EZ(I-1,J,K))*0.5                                                    1122     266
332  00B134I          EZN=(EZ(I,J+1,K)+EZ(I,J,K))*0.5                                                    1123     267
333  00B190I          EZS=(EZ(I,J,K)+EZ(I,J-1,K))*0.5                                                    1124     268
334  00B1ECI          EZT=(EZ(I,J,K+1)+EZ(I,J,K))*0.5                                                    1125     269
335  00B246I          EZB=(EZ(I,J,K)+EZ(I,J,K-1))*0.5                                                    1126     270
336  00B2A0I          SZE=(SZ(I+1,J,K)+SZ(I,J,K))*0.5                                                    1127     271
337  00B2FCI          SZW=(SZ(I,J,K)+SZ(I-1,J,K))*0.5                                                    1128     272
338  00B358I          SZN=(SZ(I,J+1,K)+SZ(I,J,K))*0.5                                                    1129     273
339  00B394I          SZS=(SZ(I,J,K)+SZ(I,J-1,K))*0.5                                                    1130     274
340  00B410I          SZT=(SZ(I,J,K+1)+SZ(I,J,K))*0.5                                                    1131     275
341  00B46AI          SZB=(SZ(I,J,K)+SZ(I,J,K-1))*0.5                                                    1132     276
342  00B4C4I          QE=GAE*(UCE*CZE+UEE*EZE+USE*SZE)                                                   1133     277
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343  0084F8I          QW=GAW*(UCW*CZW+UEW*EZW+USW*SZW)                                    1134     278
344  00852CI          QN=GAN*(UCN*CZN+UEN*EZN+USN*SZN)                                    1135     279
345  008560I          QS=GAS*(UCS*CZS+UES*EZS+USS*SZS)                                    1136     280
346  008594I          QT=GAT*(UCT*CZT+UET*EZT+UST*SZT)                                    1137     281
347  0085C8I          QB=GAB*(UCB*CZB+UEB*EZB+USB*SZB)                                    1138     282
348  0085FCI          SOC1=CX(I,J,K)*(QE-QW)+EX(I,J,K)*(QN-QS)+SX(I,J,K)*(QT-QB)          1139     283
349  008630I          QE=GAE*(VCE*CZE+VEE*EZE+VSE*SZE)                                    1140     284
350  008664I          QW=GAW*(VCW*CZW+VEW*EZW+VSW*SZW)                                    1141     285
351  008704I          QN=GAN*(VCN*CZN+VEN*EZN+VSN*SZN)                                    1142     286
352  008738I          QS=GAS*(VCS*CZS+VES*EZS+VSS*SZS)                                    1143     287
353  00876CI          QT=GAT*(VCT*CZT+VET*EZT+VST*SZT)                                    1144     288
354  0087A0I          QB=GAB*(VCB*CZB+VEB*EZB+VSB*SZB)                                    1145     289
355  0087D4I          SOC2=CY(I,J,K)*(QE-QW)+EY(I,J,K)*(QN-QS)+SY(I,J,K)*(QT-QB)          1146     290
356  008874I          QE=GAE*(WCE*CZE+WEE*EZE+WSE*SZE)                                    1147     291
357  0088A8I          QW=GAW*(WCW*CZW+WEW*EZW+WSW*SZW)                                    1148     292
358  0088DCI          QN=GAN*(WCN*CZN+WEN*EZN+WSN*SZN)                                    1149     293
359  008910I          QS=GAS*(WCS*CZS+WES*EZS+WSS*SZS)                                    1150     294
360  008944I          QT=GAT*(WCT*CZT+WET*EZT+WST*SZT)                                    1151     295
361  008978I          QB=GAB*(WCB*CZB+WEB*EZB+WSB*SZB)                                    1152     296
362  0089ACI          SOC3=CZ(I,J,K)*(QE-QW)+EZ(I,J,K)*(QN-QS)+SZ(I,J,K)*(QT-QB)          1153     297
363  0089A4CI         DW(I,J,K)=DW(I,J,K)+SOC1+SOC2+SOC3                                  1154     298
364  0089AAI          SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                      1155     299
365  0089B1CI         SP(I,J,K)=SPK(I,J,K)*TJO(I,J,K)                                     1156     300
366  0089B8EI       15  CONTINUE                                                          1157     301
367  0089BDCI           GO TO 6C                                                          1158     302
368               C-----V-SOURCE                                                        1159
369  0089E2I        2   CONTINUE                                                        1160     303
370  0089E2I            DO 25 I=IS,IT                                                   1161     304
371  0089FAI            DO 25 J=JS,JT                                                   1162     305
372  008C12I            DO 25 K=KS,KT                                                   1163     306
373  008C2AI            SU(I,J,K)=(DV(I,J,K)+APC(I,J,K)*FO(I,J,K))*TJO(I,J,K)           1164     307
374  008CEEI        25  CONTINUE                                                        1165     308
375  008D36I            GO TO 6C                                                        1166     309
376               C-----W-SOURCE                                                        1167
377  008D3CI        3   CONTINUE                                                        1168     310
378  008D3CI            DO 35 I=IS,IT                                                   1169     311
379  008D54I            DO 35 J=JS,JT                                                   1170     312
380  008D6CI            DO 35 K=KS,KT                                                   1171     313
381  008D84I            SU(I,J,K)=(DW(I,J,K)+APO(I,J,K)*FO(I,J,K))*TJO(I,J,K)           1172     314
382  008E48I        35  CONTINUE                                                        1173     315
383  008E90I            GO TO 6C                                                        1174     316
384  -                 C-----TM-SOURCE                                                1175
385  008E96I        4   CONTINUE                                                        1176     317
386  008E96I            GO TO 6C                                                        1177     318
387               C-----K-SOURCE                                                        1178
388  008E9CI        5   CONTINUE                                                        1179     319
389  008E9CI            DO 55 I=IS,IT                                                   1180     320
390  008EB4I            DO 55 J=JS,JT                                                   1181     321
391  008ECCI            DO 55 K=KS,KT                                                   1182     322
392  008EE4I            SU(I,J,K)=GEN(I,J,K)+PP(I,J,K)+APO(I,J,K)*FO(I,J,K)             1183     323
393  008FAAI            P1=DEN(I,J,K)**2                                                1184     324
394  008FDEI            SP(I,J,K)=SPK(I,J,K)-CMU*P1*F(I,J,K)/VISE(I,J,K)                1185     325
395  00C08AI            SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                  1186     326
396  00C0FCI            SP(I,J,K)=SP(I,J,K)*TJO(I,J,K)                                  1187     327
397  00C16EI        55  CONTINUE                                                        1188     328
398  00C186I            GO TO 6C                                                        1189     329
399               C-----E-SOURCE                                                        1190
```

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400  00C18CI      6   CONTINUE                                                          1191     330
401  00C18CI          DO 65 I=IS,IT                                                     1192     331
402  00C104I          DO 65 J=JS,JT                                                     1193     332
403  00C1ECI          DO 65 K=KS,KT                                                     1194     333
404  00C204I          P1=DEN(I,J,K)**2                                                  1195     334
405  00C238I          SU(I,J,K)=SU(I,J,K)+C1*CMU*GEN(I,J,K)*P1*DK(I,J,K)/               1196     335
406                  1          VISE(I,J,K)+APO(I,J,K)*FO(I,J,K)                      1197
407  00C35CI          TMDK=DK(I,J,K)+SMNUM                                              1198     336
408  00C38EI          SP(I,J,K)=SPK(I,J,K)-C2*DEN(I,J,K)*F(I,J,K)/TMDK                  1199     337
409  00C43AI          SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                    1200     338
410  00C4ACI          SP(I,J,K)=SP(I,J,K)*TJO(I,J,K)                                    1201     339
411  00C51EI      65  CONTINUE                                                          1202     340
412  00C566I          GO TO 60                                                          1203     341
413               C-----PF-SOURCE                                                      1204
414  00C56CI      7   CONTINUE                                                          1205     342
415  00C56CI          DO 75 I=IS,IT                                                     1206     343
416  00C584I          DO 75 J=JS,JT                                                     1207     344
417  00C59CI          DO 75 K=KS,KT                                                     1208     345
418  00C5B4I          SU(I,J,K)=SU(I,J,K)+0.85*GEN(I,J,K)+APO(I,J,K)*FO(I,J,K)          1209     346
419  00C680I          TMDK=DK(I,J,K)+SMNUM                                              1210     347
420  00C682I          SP(I,J,K)=SPK(I,J,K)-DE(I,J,K)/TMDK                               1211     348
421  00C72CI          SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                    1212     349
422  00C79EI          SP(I,J,K)=SP(I,J,K)*TJO(I,J,K)                                    1213     350
423  00C810I      75  CONTINUE                                                          1214     351
424  00C858I      60  CONTINUE                                                          1215     352
425               C-----MODIFY WALL BOUNDARY CONDITIONS THRU WALL FUNCTIONS             1216
426  00C858I          IF(IG .NE. 2) GO TO 410                                           1217     353
427  00C86EI          CALL BOUNC(IE,F)                                                  1218     355
428  00C89CI  410    CONTINUE                                                           1219     356
429               C-----SET SYMMETRIC, CYCLIC AND EXIT LINK COEFF.                     1220
430  00C89CI          CALL SYMOUT(2,IE,IS,IT,JS,JT,KS,KT)                               1221     357
431               C-----LINK COEFF. ASSEMBLY AND BLOCKAGES                             1222
432  00C8D8I          DO 500 I=IS,IT                                                    1223     358
433  00C8F0I          DO 500 J=JS,JT                                                    1224     359
434  00C908I          DO 500 K=KS,KT                                                    1225     360
435  00C920I          F1(I,J,K)=F(I,J,K)                                                1226     361
436  00C972I          ANAB=AE(I,J,K)+AW(I,J,K)+AN(I,J,K)+AS(I,J,K)+AT(I,J,K)+           1227     362
437                  1     AB(I,J,K)+APO(I,J,K)                                        1228
438  00CA82I          AP(I,J,K)=ANAB-SP(I,J,K)                                          1229     363
439  00CAD4I          PDUV=1.0                                                          1230     364
440  00CAE0I          IF(MC(I,J,K) .LT. 1) GO TO 530                                    1231     365
441  00CB16I          AP(I,J,K)=ALF                                                     1232     367
442  00CB42I          AN(I,J,K)=0.0                                                     1233     368
443  00CB6EI          AS(I,J,K)=0.0                                                     1234     369
444  00CB9AI          AE(I,J,K)=0.0                                                     1235     370
445  00CBC6I          AW(I,J,K)=0.0                                                     1236     371
446  00CBF2I          AT(I,J,K)=0.0                                                     1237     372
447  00CC1EI          AB(I,J,K)=0.0                                                     1238     373
448  00CC4AI          SU(I,J,K)=F(I,J,K)                                                1239     374
449  00CC9CI          PDUV=0.0                                                          1240     375
450  00CCA8I  530    CONTINUE                                                           1241     376
451               C-----UNDER-RELAXATION                                               1242
452  00CCA8I          P1=1.2*AP(I,J,K)                                                  1243     377
453  00CCDAI          AP(I,J,K)=AP(I,J,K)/ALF                                           1244     378
454  00CD2CI          SU(I,J,K)=SU(I,J,K)+PDUV*(1.0-ALF)*AP(I,J,K)*F(I,J,K)             1245     379
455  00CDDCI          IF(IE .EQ. 1) DU(I,J,K)=TJO(I,J,K)*PDUV/(P1-ANAB)                 1246     380
456  00CE4CI  500    CONTINUE                                                           1247     382
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457                C-----LINEAR EQUATIONS SLOVER                                                    1248
458   00CE94I            DO 550 I=1,ISWF                                                          1249      383
459   00CEA8I     550    CALL LINERX(1,IS,JS,KS,IT,JT,KT,F)                                       1250      384
460                C-----CALCULATE MAXIMUM CORRECTION OF CURRENT ITERATION                         1251
461   00CF0CI            DO 555 I=IS,IT                                                           1252      385
462   00CF24I            DO 555 J=JS,JT                                                           1253      386
463   00CF3CI            DO 555 K=KS,KT                                                           1254      387
464   00CF54I            P1=ABS(F(I,J,K)-F1(I,J,K))                                               1255      388
465   00CFCAI            ERRF=AMAX1(ERRF,P1)                                                      1256      389
466   00CFEAI     555    CONTINUE                                                                 1257      390
467   00D032I            RETURN                                                                   1258      391
468                C-------------------------------------------------------------------------    1259
469                C-----P EQUATION                                                                1260
470   00D03AI       10   CONTINUE                                                                 1261      392
471   00D03AI            IS=2                                                                     1262      393
472   00D042I            IT=L                                                                     1263      394
473   00D04EI            JS=2                                                                     1264      395
474   00D056I            JT=M                                                                     1265      396
475   00D062I            KS=2                                                                     1266      397
476   00D06AI            KT=N                                                                     1267      398
477                C-----SOURCE AT PX LOCATIONS                                                    1268
478   00D076I            DO 310 I=IS,IT                                                           1269      399
479   00D08EI            DO 310 J=JS,JT                                                           1270      400
480   00D0A6I            DO 310 K=KS,KT                                                           1271      401
481   00D0BEI            F(I,J,K)=0.0                                                             1272      402
482   00D0F0I            SUK(I,J,K)=0.0                                                           1273      403
483   00D11CI            SPK(I,J,K)=0.0                                                           1274      404
484   00D148I            SP(I,J,K)=0.0                                                            1275      405
485   00D174I            DENE=DEN(I,J,K)                                                          1276      406
486   00D1A0I            DENW=DEN(I-1,J,K)                                                        1277      407
487   00D1D0I            DENN=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J+1,K)+DEN(I-1,J+1,K))          1278      408
488   00D284I            DENS=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J-1,K)+DEN(I-1,J-1,K))          1279      409
489   00D338I            DENT=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J,K+1)+DEN(I-1,J,K+1))          1280      410
490   00D3E8I            DENB=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J,K-1)+DEN(I-1,J,K-1))          1281      411
491   00D498I            UE=U(I,J,K)                                                              1282      412
492   00D4C4I            UW=U(I-1,J,K)                                                            1283      413
493   00D4F4I            UN=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J+1,K)+U(I-1,J+1,K))                    1284      414
494   00D5A8I            US=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J-1,K)+U(I-1,J-1,K))                    1285      415
495   00D65CI            UT=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J,K+1)+U(I-1,J,K+1))                    1286      416
496   00D70CI            UB=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J,K-1)+U(I-1,J,K-1))                    1287      417
497   00D7BCI            VE=V(I,J,K)                                                              1288      418
498   00D7E8I            VW=V(I-1,J,K)                                                            1289      419
499   00D818I            VN=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J+1,K)+V(I-1,J+1,K))                    1290      420
500   00D8CCI            VS=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J-1,K)+V(I-1,J-1,K))                    1291      421
501   00D980I            VT=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J,K+1)+V(I-1,J,K+1))                    1292      422
502   00DA30I            VB=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J,K-1)+V(I-1,J,K-1))                    1293      423
503   00DAE0I            WE=W(I,J,K)                                                              1294      424
504   00DB0CI            WW=W(I-1,J,K)                                                            1295      425
505   00DB3CI            WN=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J+1,K)+W(I-1,J+1,K))                    1296      426
506   00DBF0I            WS=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J-1,K)+W(I-1,J-1,K))                    1297      427
507   00DCA4I            WT=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J,K+1)+W(I-1,J,K+1))                    1298      428
508   00DD54I            WB=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J,K-1)+W(I-1,J,K-1))                    1299      429
509   00DE04I            CXQ=0.5*(CX(I,J,K)+CX(I-1,J,K))                                          1300      430
510   00DE60I            EXQ=0.5*(EX(I,J,K)+EX(I-1,J,K))                                          1301      431
511   00DE6CI            SXQ=0.5*(SX(I,J,K)+SX(I-1,J,K))                                          1302      432
512   00DF18I            CYQ=0.5*(CY(I,J,K)+CY(I-1,J,K))                                          1303      433
513   00DF74I            EYQ=0.5*(EY(I,J,K)+EY(I-1,J,K))                                          1304      434
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514  000F90I          SYQ=0.5*(SY(I,J,K)+SY(I-1,J,K))                                                                    1305     435
515  00E02CI          CZQ=0.5*(CZ(I,J,K)+CZ(I-1,J,K))                                                                    1306     436
516  00E088I          EZQ=0.5*(EZ(I,J,K)+EZ(I-1,J,K))                                                                    1307     437
517  00E0E4I          SZQ=0.5*(SZ(I,J,K)+SZ(I-1,J,K))                                                                    1308     438
518  00E140I          CE=CENE*(UE*CXQ+VE*CYQ+WE*CZQ)                                                                     1309     439
519  00E174I          CW=CENW*(UW*CXQ+VW*CYQ+WW*CZQ)                                                                     1310     440
520  00E1A8I          CN=CENN*(UN*EXQ+VN*EYQ+WN*EZQ)                                                                     1311     441
521  00E1DCI          CS=CENS*(US*EXQ+VS*EYQ+WS*EZQ)                                                                     1312     442
522  00E210I          CT=CENT*(UT*SXQ+VT*SYQ+WT*SZQ)                                                                     1313     443
523  00E244I          CB=CENB*(UB*SXQ+VB*SYQ+WB*SZQ)                                                                     1314     444
524  00E278I          SUK(I,J,K)=-(CE-CW+CN-CS+CT-CB)*(TJO(I-1,J,K)+TJO(I,J,K))*0.5                                      1315     445
525  00E322I    310   CONTINUE                                                                                           1316     446
526               C-----SOURCE AT PY LOCATIONS                                                                         1317
527  00E36AI          DO 311 I=IS,IT                                                                                     1318     447
528  00E382I          DO 311 J=JS,JT                                                                                     1319     448
529  00E39AI          DO 311 K=KS,KT                                                                                     1320     449
530  00E3B2I          DENN=DEN(I,J,K)                                                                                    1321     450
531  00E3DEI          DENS=DEN(I,J-1,K)                                                                                  1322     451
532  00E40EI          DENE=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I+1,J,K)+DEN(I+1,J-1,K))                                    1323     452
533  00E4C2I          DENW=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I-1,J,K)+DEN(I-1,J-1,K))                                    1324     453
534  00E576I          DENT=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K+1)+DEN(I,J-1,K+1))                                    1325     454
535  00E626I          DENB=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K-1)+DEN(I,J-1,K-1))                                    1326     455
536  00E6D6I          UN=U(I,J,K)                                                                                        1327     456
537  00E702I          US=U(I,J-1,K)                                                                                      1328     457
538  00E732I          UE=0.25*(U(I,J,K)+U(I,J-1,K)+U(I+1,J,K)+U(I+1,J-1,K))                                              1329     458
539  00E7E6I          UW=0.25*(U(I,J,K)+U(I,J-1,K)+U(I-1,J,K)+U(I-1,J-1,K))                                              1330     459
540  00E89AI          UT=0.25*(U(I,J,K)+U(I,J-1,K)+U(I,J,K+1)+U(I,J-1,K+1))                                              1331     460
541  00E94AI          UB=0.25*(U(I,J,K)+U(I,J-1,K)+U(I,J,K-1)+U(I,J-1,K-1))                                              1332     461
542  00E9FAI          VN=V(I,J,K)                                                                                        1333     462
543  00EA26I          VS=V(I,J-1,K)                                                                                      1334     463
544  00EA56I          VE=0.25*(V(I,J,K)+V(I,J-1,K)+V(I+1,J,K)+V(I+1,J-1,K))                                              1335     464
545  00EB0AI          VW=0.25*(V(I,J,K)+V(I,J-1,K)+V(I-1,J,K)+V(I-1,J-1,K))                                              1336     465
546  00EB3EI          VT=0.25*(V(I,J,K)+V(I,J-1,K)+V(I,J,K+1)+V(I,J-1,K+1))                                              1337     466
547  00EC6EI          VB=0.25*(V(I,J,K)+V(I,J-1,K)+V(I,J,K-1)+V(I,J-1,K-1))                                              1338     467
548  00ED1EI          WN=W(I,J,K)                                                                                        1339     468
549  00ED4AI          WS=W(I,J-1,K)                                                                                      1340     469
550  00ED7AI          WE=0.25*(W(I,J,K)+W(I,J-1,K)+W(I+1,J,K)+W(I+1,J-1,K))                                              1341     470
551  00EE2EI          WW=0.25*(W(I,J,K)+W(I,J-1,K)+W(I-1,J,K)+W(I-1,J-1,K))                                              1342     471
552  00EEE2I          WT=0.25*(W(I,J,K)+W(I,J-1,K)+W(I,J,K+1)+W(I,J-1,K+1))                                              1343     472
553  00EF92I          WB=0.25*(W(I,J,K)+W(I,J-1,K)+W(I,J,K-1)+W(I,J-1,K-1))                                              1344     473
554  00F042I          CXQ=0.5*(CX(I,J,K)+CX(I,J-1,K))                                                                    1345     474
555  00F09EI          EXQ=0.5*(EX(I,J,K)+EX(I,J-1,K))                                                                    1346     475
556  00F0FAI          SXQ=0.5*(SX(I,J,K)+SX(I,J-1,K))                                                                    1347     476
557  00F156I          CYQ=0.5*(CY(I,J,K)+CY(I,J-1,K))                                                                    1348     477
558  00F1B2I          EYQ=0.5*(EY(I,J,K)+EY(I,J-1,K))                                                                    1349     478
559  00F20EI          SYQ=0.5*(SY(I,J,K)+SY(I,J-1,K))                                                                    1350     479
560  00F26AI          CZQ=0.5*(CZ(I,J,K)+CZ(I,J-1,K))                                                                    1351     480
561  00F2C6I          EZQ=0.5*(EZ(I,J,K)+EZ(I,J-1,K))                                                                    1352     481
562  00F322I          SZQ=0.5*(SZ(I,J,K)+SZ(I,J-1,K))                                                                    1353     482
563  00F37EI          CE=CENE*(UE*CXQ+VE*CYQ+WE*CZQ)                                                                     1354     483
564  00F3B2I          CW=CENW*(UW*CXQ+VW*CYQ+WW*CZQ)                                                                     1355     484
565  00F3E6I          CN=CENN*(UN*EXQ+VN*EYQ+WN*EZQ)                                                                     1356     485
566  00F41AI          CS=CENS*(US*EXQ+VS*EYQ+WS*EZQ)                                                                     1357     486
567  00F44EI          CT=CENT*(UT*SXQ+VT*SYQ+WT*SZQ)                                                                     1358     487
568  00F482I          CB=CENB*(UB*SXQ+VB*SYQ+WB*SZQ)                                                                     1359     488
569  00F486I          SPK(I,J,K)=-(CE-CW+CN-CS+CT-CB)*(TJO(I,J-1,K)+TJO(I,J,K))*0.5                                      1360     489
570  00F560I    311   CONTINUE                                                                                           1361     490
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571            C-----SOURCE AT PZ LOCATIONS                                                                    1362
572   00F5A8I          DO 312 I=IS,IT                                                                          1363      491
573   00F5C0I          DO 312 J=JS,JT                                                                          1364      492
574   00F5D8I          DO 312 K=KS,KT                                                                          1365      493
575   00F5F0I          DENT=DEN(I,J,K)                                                                         1366      494
576   00F61CI          DENS=DEN(I,J,K-1)                                                                       1367      495
577   00F64AI          DENN=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I,J+1,K)+DEN(I,J+1,K-1))                         1368      496
578   00F6FAI          DENS=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I,J-1,K)+DEN(I,J-1,K-1))                         1369      497
579   00F7AAI          DENE=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I+1,J,K)+DEN(I+1,J,K-1))                         1370      498
580   00F85AI          DENW=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I-1,J,K)+DEN(I-1,J,K-1))                         1371      499
581   00F90AI          UT=U(I,J,K)                                                                             1372      500
582   00F936I          UB=U(I,J,K-1)                                                                           1373      501
583   00F964I          UN=0.25*(U(I,J,K)+U(I,J,K-1)+U(I,J+1,K)+U(I,J+1,K-1))                                   1374      502
584   00FA14I          US=0.25*(U(I,J,K)+U(I,J,K-1)+U(I,J-1,K)+U(I,J-1,K-1))                                   1375      503
585   00FAC4I          UE=0.25*(U(I,J,K)+U(I,J,K-1)+U(I+1,J,K)+U(I+1,J,K-1))                                   1376      504
586   00FB74I          UW=0.25*(U(I,J,K)+U(I,J,K-1)+U(I-1,J,K)+U(I-1,J,K-1))                                   1377      505
587   00FC24I          VT=V(I,J,K)                                                                             1378      506
588   00FC50I          VB=V(I,J,K-1)                                                                           1379      507
589   00FC7EI          VN=0.25*(V(I,J,K)+V(I,J,K-1)+V(I,J+1,K)+V(I,J+1,K-1))                                   1380      508
590   00FD2EI          VS=0.25*(V(I,J,K)+V(I,J,K-1)+V(I,J-1,K)+V(I,J-1,K-1))                                   1381      509
591   00FDDEI          VE=0.25*(V(I,J,K)+V(I,J,K-1)+V(I+1,J,K)+V(I+1,J,K-1))                                   1382      510
592   00FE8EI          VW=0.25*(V(I,J,K)+V(I,J,K-1)+V(I-1,J,K)+V(I-1,J,K-1))                                   1383      511
593   00FF3EI          WT=W(I,J,K)                                                                             1384      512
594   00FF6AI          WB=W(I,J,K-1)                                                                           1385      513
595   00FF98I          WN=0.25*(W(I,J,K)+W(I,J,K-1)+W(I,J+1,K)+W(I,J+1,K-1))                                   1386      514
596   0100A8I          WS=0.25*(W(I,J,K)+W(I,J,K-1)+W(I,J-1,K)+W(I,J-1,K-1))                                   1387      515
597   0100F8I          WE=0.25*(W(I,J,K)+W(I,J,K-1)+W(I+1,J,K)+W(I+1,J,K-1))                                   1388      516
598   0101A8I          WW=0.25*(W(I,J,K)+W(I,J,K-1)+W(I-1,J,K)+W(I-1,J,K-1))                                   1389      517
599   010258I          CXQ=0.5*(CX(I,J,K)+CX(I,J,K-1))                                                         1390      518
600   0102B2I          EXQ=0.5*(EX(I,J,K)+EX(I,J,K-1))                                                         1391      519
601   01030CI          SXQ=0.5*(SX(I,J,K)+SX(I,J,K-1))                                                         1392      520
602   010366I          CYQ=0.5*(CY(I,J,K)+CY(I,J,K-1))                                                         1393      521
603   0103C0I          EYQ=0.5*(EY(I,J,K)+EY(I,J,K-1))                                                         1394      522
604   01041AI          SYQ=0.5*(SY(I,J,K)+SY(I,J,K-1))                                                         1395      523
605   010474I          CZQ=0.5*(CZ(I,J,K)+CZ(I,J,K-1))                                                         1396      524
606   0104CEI          EZQ=0.5*(EZ(I,J,K)+EZ(I,J,K-1))                                                         1397      525
607   010528I          SZQ=0.5*(SZ(I,J,K)+SZ(I,J,K-1))                                                         1398      526
608   0105B2I          CE=DENE*(UE*CXQ+VE*CYQ+WE*CZQ)                                                          1399      527
609   0105E6I          CW=DENW*(UW*CXQ+VW*CYQ+WW*CZQ)                                                          1400      528
610   01061AI          CN=DENN*(UN*EXQ+VN*EYQ+WN*EZQ)                                                          1401      529
611   01064EI          CS=DENS*(US*EXQ+VS*EYQ+WS*EZQ)                                                          1402      530
612   010682I          CT=DENT*(UT*SXQ+VT*SYQ+WT*SZQ)                                                          1403      531
613   0106B6I          CB=DENB*(UB*SXQ+VB*SYQ+WB*SZQ)                                                          1404      532
614   0106EAI          SP(I,J,K)=-(CE-CW+CN-CS+CT-CB)*(TJO(I,J,K-1)+TJO(I,J,K))*0.5                            1405      533
615   010762I    312   CONTINUE                                                                                1406      534
616            C-----LINK COEFF. AND SOURCE TERM AT CELL CENTER                                               1407
617   0107AAI          DO 100 I=IS,IT                                                                          1408      535
618   0107C2I          DO 100 J=JS,JT                                                                          1409      536
619   0107DAI          DO 100 K=KS,KT                                                                          1410      537
620   0107F2I          II=I-1                                                                                  1411      538
621   010800I          JJ=J-1                                                                                  1412      539
622   01080EI          KK=K-1                                                                                  1413      540
623   01081CI          GAE=(DU(I,J,K)+DU(I,JJ,K)+DU(I,J,KK)+DU(I,JJ,KK))*0.25                                  1414      541
624   0108C0I          GAW=(DU(II,J,K)+DU(II,JJ,K)+DU(II,J,KK)+DU(II,JJ,KK))*0.25                              1415      542
625   010964I          GAN=(DU(I,J,K)+DU(II,J,K)+DU(I,J,KK)+DU(II,J,KK))*0.25                                  1416      543
626   0109A8I          GAS=(DU(I,JJ,K)+DU(II,JJ,K)+DU(I,JJ,KK)+DU(II,JJ,KK))*0.25                              1417      544
627   010AACI          GAT=(DU(I,J,K)+DU(II,J,K)+DU(I,JJ,K)+DU(II,JJ,K))*0.25                                  1418      545
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628  010B50I         GAB=(DU(I,J,KK)+DU(II,J,KK)+DU(I,JJ,KK)+DU(II,JJ,KK))*0.25                1419     546
629  010BF4I         TXXEQ=(TXXE(I,J,K)+TXXE(II,JJ,KK))*0.5                                   1420     547
630  010C4CI         TXXWQ=(TXXW(I,J,K)+TXXW(II,JJ,KK))*0.5                                   1421     548
631  010CA4I         TYYNQ=(TYYN(I,J,K)+TYYN(II,JJ,KK))*0.5                                   1422     549
632  010CFCI         TYYSQ=(TYYS(I,J,K)+TYYS(II,JJ,KK))*0.5                                   1423     550
633  010D54I         TZZTQ=(TZZT(I,J,K)+TZZT(II,JJ,KK))*0.5                                   1424     551
634  010DACI         TZZBQ=(TZZB(I,J,K)+TZZB(II,JJ,KK))*0.5                                   1425     552
635  010E04I         TQ=(TJC(I,J,K)+TJC(II,JJ,KK))*0.5                                        1426     553
636  010E5CI         AE(I,J,K)=TQ*(GAE*TXXEQ)                                                  1427     554
637  010E94I         AW(I,J,K)=TQ*(GAW*TXXWQ)                                                  1428     555
638  010ECCI         AN(I,J,K)=TQ*(GAN*TYYNQ)                                                  1429     556
639  010F04I         AS(I,J,K)=TQ*(GAS*TYYSQ)                                                  1430     557
640  010F3CI         AT(I,J,K)=TQ*(GAT*TZZTQ)                                                  1431     558
641  010F74I         AB(I,J,K)=TQ*(GAB*TZZBQ)                                                  1432     559
642                C-----P' SOURCE                                                             1433
643  010FACI         P1=1.0                                                                    1434     560
644  010FB8I         P2=1.0                                                                    1435     561
645  010FC4I         P3=1.0                                                                    1436     562
646  010FD0I         P4=1.0                                                                    1437     563
647  010FDCI         P5=1.0                                                                    1438     564
648  010FE8I         P6=1.0                                                                    1439     565
649  010FF4I         IF(GAN .EQ. 0.0) P1=0.0                                                   1440     566
650  011012I         IF(GAS .EQ. 0.0) P2=0.0                                                   1441     568
651  011030I         IF(GAE .EQ. 0.0) P3=0.0                                                   1442     570
652  01104EI         IF(GAW .EQ. 0.0) P4=0.0                                                   1443     572
653  01106CI         IF(GAT .EQ. 0.0) P5=0.0                                                   1444     574
654  01108AI         IF(GAB .EQ. 0.0) P6=0.0                                                   1445     576
655  0110A8I         PQ=0.5*((P1+P2)*(P3+P4+P5+P6)+(P3+P4)*(P1+P2+P5+P6)+                      1446     578
656                1          (P5+P6)*(P1+P2+P3+P4))                                         1447
657  0110C6I         IF(PQ .EQ. 0.0) PQ=1.0                                                    1448     579
658  0110E4I         SU(I,J,K)=(P1*P5*SUK(I,J,K)+P5*P2*SUK(I,J-1,K)+                           1449     581
659                1           P2*P6*SUK(I,J-1,K-1)+P6*P1*SUK(I,J,K-1)+                      1450
660                2           P3*P5*SPK(I,J,K)+P5*P4*SPK(I-1,J,K)+                          1451
661                3           P4*P6*SPK(I-1,J,K-1)+P6*P3*SPK(I,J,K-1)+                      1452
662                4           P1*P3*SP(I,J,K)+P3*P2*SP(I,J-1,K)+                            1453
663                5           P2*P4*SP(I-1,J-1,K)+P4*P1*SP(I-1,J,K))/PQ                     1454
664  011102I      100  CONTINUE                                                              1455     582
665                C-----LINK COEFF. ASSEMBLY AND BLOCKAGES                                  1456
666  011120I         DO 870 I=IS,IT                                                            1457     583
667  01113EI         DO 870 J=JS,JT                                                            1458     584
668  01115CI         DO 870 K=KS,KT                                                            1459     585
669  01117AI         ANAB=AE(I,J,K)+AW(I,J,K)+AN(I,J,K)+AS(I,J,K)+AT(I,J,K)+AB(I,J,K)          1460     586
670  011198I         AP(I,J,K)=ANAB                                                            1461     587
671  0111B6I         IF(MC(I,J,K) .LT. 2) GO TO 870                                            1462     588
672  0111D4I         AP(I,J,K)=1.0                                                             1463     590
673  0111F2I         AE(I,J,K)=0.0                                                             1464     591
674  011210I         AW(I,J,K)=0.0                                                             1465     592
675  01122EI         AN(I,J,K)=0.0                                                             1466     593
676  01124CI         AS(I,J,K)=0.0                                                             1467     594
677  01126AI         AT(I,J,K)=0.0                                                             1468     595
678  011288I         AB(I,J,K)=0.0                                                             1469     596
679  0112A6I         SU(I,J,K)=0.0                                                             1470     597
680  0112C4I      870  CONTINUE                                                              1471     598
681                C-----LINEAR EQUATIONS SOLVER                                             1472
682  0112E2I         CALL LINERX(2,IS,JS,KS,IT,JT,KT,F)                                        1473     599
683  011300I         DO 850 I=1,ISWF                                                           1474     600
684  01131EI      850  CALL LINERX(1,IS,JS,KS,IT,JT,KT,F)                                    1475     601
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685                C-----PRESSURE AND VELOCITIES CORRECTIONS                                    1476
686   0118541          PPREF=F(2,2,2)                                                          1477      602
687   0118641          DO 880 I=IS,IT                                                          1478      603
688   01187CI          DO 880 J=JS,JT                                                          1479      604
689   0118941          DO 880 K=KS,KT                                                          1480      605
690   0118AC1          IF(MC(I,J,K) .GE. 2) GO TO 880                                          1481      606
691   0118E21          P1=F(I,J,K)-PPREF                                                       1482      608
692   01191AI          P(I,J,K)=P(I,J,K)+ALP*P1                                                1483      609
693   0119721          ERRF=AMAX1(ERRF,ABS(P1))                                                1484      610
694   0119AEI    880   CONTINUE                                                                1485      611
695   0119F61          DO 600 I=2,LT                                                           1486      612
696   011A0AI          DO 600 J=2,MT                                                           1487      613
697   011A1EI          DO 600 K=2,NT                                                           1488      614
698   011A321          IF(MC(I,J,K) .GE. 1) GO TO 600                                          1489      615
699   011A681          II=I+1                                                                  1490      617
700   011A761          JJ=J+1                                                                  1491      618
701   011A841          KK=K+1                                                                  1492      619
702   011A921          PE=(F(II,J,K)+F(II,JJ,K)+F(II,J,KK)+F(II,JJ,KK))*0.25                   1493      620
703   011B4EI          PW=(F(I,J,K)+F(I,JJ,K)+F(I,J,KK)+F(I,JJ,KK))*0.25                       1494      621
704   011C0AI          PN=(F(I,JJ,K)+F(II,JJ,K)+F(I,JJ,KK)+F(II,JJ,KK))*0.25                   1495      622
705   011CC61          PS=(F(I,J,K)+F(II,J,K)+F(I,J,KK)+F(II,J,KK))*0.25                       1496      623
706   011D821          PT=(F(I,J,KK)+F(II,J,KK)+F(I,JJ,KK)+F(II,JJ,KK))*0.25                   1497      624
707   011E3EI          PB=(F(I,J,K)+F(II,J,K)+F(I,JJ,K)+F(II,JJ,K))*0.25                       1498      625
708   011EFAI          PCXI=PE-PW                                                              1499      626
709   011F0C1          PEDA=PN-PS                                                              1500      627
710   011F1EI          PSCI=PT-PB                                                              1501      628
711   011F301          CXQ=CX(I,J,K)                                                           1502      629
712   011F5C1          EXQ=EX(I,J,K)                                                           1503      630
713   011F881          SXQ=SX(I,J,K)                                                           1504      631
714   011F941          CYQ=CY(I,J,K)                                                           1505      632
715   011FE01          EYQ=EY(I,J,K)                                                           1506      633
716   01200C1          SYQ=SY(I,J,K)                                                           1507      634
717   0120381          CZQ=CZ(I,J,K)                                                           1508      635
718   0120641          EZQ=EZ(I,J,K)                                                           1509      636
719   0120901          SZQ=SZ(I,J,K)                                                           1510      637
720   0120BC1          PXX=PCXI*CXQ+PEDA*EXQ+PSCI*SXQ                                          1511      638
721   0120EAI          PYY=PCXI*CYQ+PEDA*EYQ+PSCI*SYQ                                          1512      639
722   0121181          PZZ=PCXI*CZQ+PEDA*EZQ+PSCI*SZQ                                          1513      640
723   0121461          U(I,J,K)=U(I,J,K)-DU(I,J,K)*PXX                                         1514      641
724   0121C01          V(I,J,K)=V(I,J,K)-DU(I,J,K)*PYY                                         1515      642
725   01223AI          W(I,J,K)=W(I,J,K)-DU(I,J,K)*PZZ                                         1516      643
726   0122841    600   CONTINUE                                                                1517      644
727   0122FC1          RETURN                                                                  1518      645
728   0123041          END                                                                     1519      646

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  SOLVEQ    02/21/86  09:58:31  TABLE SPACE:  18 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 203 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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 1  000000I          SUBROUTINE LINERX(ISOL,IS,JS,KS,LT,MT,NT,F)                    1520
 2  000004I          DIMENSION A(20),B(20),C(20),D(20),F(21,18,10),BB(21,18,10)     1521
 3  000004I          COMMON                                                         1522
 4                  1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),  1523
 5                  2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),      1524
 6                  3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)       1525
 7               C-----LINEAR EQUATIONS SOLVERS                                     1526
 8  000004I          GO TO (1,2),ISOL                                               1527      1
 9               C-----LINE-RELAXATION USING TOMA                                   1528
10  003C86I        1   CONTINUE                                                     1529      2
11  003C86I          DO 10 I=IS,LT                                                  1530      3
12  003C9EI          DO 10 J=JS,MT                                                  1531      4
13  003C86I          DO 10 K=KS,NT                                                  1532      5
14  003CCEI          BB(I,J,K)=AE(I,J,K)*F(I+1,J,K)+AN(I,J,K)*F(I,J+1,K)+           1533      6
15                  1          SU(I,J,K)                                           1534
16  003DC8I       10 CONTINUE                                                     1535      7
17  003E10I          A(KS-1)=0.0                                                    1536      8
18  003E24I          DO 100 I=IS,LT                                                 1537      9
19  003E3CI          DO 100 J=JS,MT                                                 1538     10
20  003E54I          C(KS-1)=F(I,J,KS-1)                                            1539     11
21  003E90I          DO 101 K=KS,NT                                                 1540     12
22  003EA8I          A(K)=AT(I,J,K)                                                 1541     13
23  003EDAI          B(K)=AB(I,J,K)                                                 1542     14
24  003F0CI          C(K)=BB(I,J,K)+AW(I,J,K)*F(I-1,J,K)+                           1543     15
25                  1     AS(I,J,K)*F(I,J-1,K)                                     1544
26  003FECI          D(K)=AP(I,J,K)                                                 1545     16
27  00401EI          TERM1=D(K)-B(K)*A(K-1)+1.E-30                                  1546     17
28  004054I          TERM=1.0/TERM1                                                 1547     18
29  004066I          A(K)=A(K)*TERM                                                 1548     19
30  004088I   101    C(K)=(C(K)+B(K)*C(K-1))*TERM                                   1549     20
31  0040E0I          DO 102 KK=KS,NT                                                1550     21
32  0040F8I          K=KS+NT-KK                                                     1551     22
33  004110I   102    F(I,J,K)=A(K)*F(I,J,K+1)+C(K)                                  1552     23
34  00419EI   100    CONTINUE                                                       1553     24
35  0041CEI          RETURN                                                         1554     25
36               C-----BLOCK CORRECTION FOR PRESSURE FIELD                          1555
37  0041D6I        2 CONTINUE                                                       1556     26
38  0041D6I          DO 201 I=IS,LT                                                 1557     27
39  0041EEI          A(I)=0.0                                                       1558     28
40  004202I          B(I)=0.0                                                       1559     29
41  004216I          C(I)=0.0                                                       1560     30
42  00422AI          D(I)=0.0                                                       1561     31
43  00423EI          DO 201 J=JS,MT                                                 1562     32
44  004256I          DO 201 K=KS,NT                                                 1563     33
45  00426EI          A(I)=A(I)-AE(I,J,K)                                            1564     34
46  004280I          B(I)=B(I)-AW(I,J,K)                                            1565     35
47  0042F2I          C(I)=C(I)+SU(I,J,K)                                            1566     36
48  004334I   201    CONTINUE                                                       1567     37
49  00437CI          DO 202 I=IS,LT                                                 1568     38
50  004394I          ANAB=-A(I)-B(I)                                                1569     39
51  00438CI   202    D(I)=ANAB*1.5                                                  1570     40
52  0043EEI          DO 203 I=IS+1,LT                                               1571     41
53  00440AI          TERM=B(I)/D(I-1)                                               1572     42
54  00442EI          D(I)=D(I)-TERM*A(I-1)                                          1573     43
55  004462I   203    C(I)=C(I)-TERM*C(I-1)                                          1574     44
56  0044AEI          F(LT,JS,KS)=C(LT)/D(LT)                                        1575     45
57  0044F6I          DO 204 II=IS+1,LT                                              1576     46
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 58  004512I          I=IS+LT-II                                                  1577      47
 59  00452AI     204  F(I,JS,KS)=(C(I)-A(I)*F(I+1,JS,KS))/D(I)                   1578      48
 60  0045CAI          DO 205 I=IS,LT                                              1579      49
 61  0045E2I          PPBLK=F(I,JS,KS)                                            1580      50
 62  004614I          DO 205 J=JS+1,MT                                            1581      51
 63  004630I          DO 205 K=KS+1,NT                                            1582      52
 64  00464CI          F(I,J,K)=PPBLK                                              1583      53
 65  00467EI     205  CONTINUE                                                    1584      54
 66  0046C6I          RETURN                                                      1585      55
 67  0046CEI          END                                                         1586      56

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  LINERX    02/21/86  09:59:43  TABLE SPACE:    3 KB
STATEMENT BUFFER:    20 LINES/1321 BYTES    STACK SPACE: 199 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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      SUBROUTINE BOUNC(IE,F)                                            1587
      DIMENSION F(21,18,10)                                             1588
      COMMON                                                            1589
     1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),            1590
     2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                     1591
     3  PP(21,18,10),W(21,18,10),TM(21,18,10)                           1592
     1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN             1593
     1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                     1594
     1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),   1595
     2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),        1596
     3 YPLN(800),TAUN(800),I3C(800),J3C(800),K3C(800),IITY(800),        1597
     4 TALW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO         1598
     1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),      1599
     2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),          1600
     3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)           1601
      COMMON                                                            1602
     1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),         1603
     2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                        1604
     3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                        1605
     3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                         1606
     1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,       1607
     2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT    1608
C-----EVALUATE WALL BOUNDARY CONDITIONS USING WALL FUNCTIONS             1609
      DO 150 III=1,IITO                                                 1610      1
      I=I3C(III)                                                        1611      2
      J=J3C(III)                                                        1612      3
      K=K3C(III)                                                        1613      4
      GO TO (1,2,3,4,5,6), IITY(III)                                    1614      5
    1 CONTINUE                                                          1615      6
C-----NORTH                                                             1616
      YP=YN(III)                                                        1617      7
      YP1=YN1(III)                                                      1618      8
      CDK=DK(I,J,K)                                                     1619      9
      CDE=DE(I,J,K)                                                     1620     10
      CDEN=DEN(I,J,K)                                                   1621     11
      CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1622     12
     1      SINZ(III),F(I,J+1,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),        1623
     2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),                1624
     3      U(I,J+1,K),V(I,J+1,K),W(I,J+1,K),U(I,J,K),V(I,J,K),        1625
     4      W(I,J,K),U(I,J-1,K),V(I,J-1,K),W(I,J-1,K),                 1626
     5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJO(I,J,K))      1627
      GO TO 150                                                         1628     13
    2 CONTINUE                                                          1629     14
C-----SOUTH                                                             1630
      YP=YN(III)                                                        1631     15
      YP1=YN1(III)                                                      1632     16
      CDK=DK(I,J,K)                                                     1633     17
      CDE=DE(I,J,K)                                                     1634     18
      CDEN=DEN(I,J,K)                                                   1635     19
      CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1636     20
     1      SINZ(III),F(I,J-1,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),        1637
     2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),                1638
     3      U(I,J-1,K),V(I,J-1,K),W(I,J-1,K),U(I,J,K),V(I,J,K),        1639
     4      W(I,J,K),U(I,J+1,K),V(I,J+1,K),W(I,J+1,K),                 1640
     5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJO(I,J,K))      1641
      GO TO 150                                                         1642     21
    3 CONTINUE                                                          1643     22
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                C-----EAST                                                                1644
000A06I                 YP=YN(III)                                                        1645        23
000A2AI                 YP1=YN1(III)                                                      1646        24
000A4EI                 CDK=DK(I,J,K)                                                     1647        25
000B2AI                 CDE=DE(I,J,K)                                                     1648        26
000B56I                 CDEN=DEN(I,J,K)                                                   1649        27
000B82I                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1650        28
                       1      SINZ(III),F(I+1,J,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1651
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1652
                       3      U(I+1,J,K),V(I+1,J,K),W(I+1,J,K),U(I,J,K),V(I,J,K),       1653
                       4      W(I,J,K),U(I-1,J,K),V(I-1,J,K),W(I-1,J,K),                1654
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1655
000FECI                 GO TO 150                                                         1656        29
000FF2I           4     CONTINUE                                                          1657        30
                C-----WEST                                                                1658
000FF2I                 YP=YN(III)                                                        1659        31
001006I                 YP1=YN1(III)                                                      1660        32
00101AI                 CDK=DK(I,J,K)                                                     1661        33
001046I                 CDE=DE(I,J,K)                                                     1662        34
001072I                 CDEN=DEN(I,J,K)                                                   1663        35
00109EI                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1664        36
                       1      SINZ(III),F(I-1,J,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1665
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1666
                       3      U(I-1,J,K),V(I-1,J,K),W(I-1,J,K),U(I,J,K),V(I,J,K),       1667
                       4      W(I,J,K),U(I+1,J,K),V(I+1,J,K),W(I+1,J,K),                1668
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1669
001508I                 GO TO 150                                                         1670        37
00150EI           5     CONTINUE                                                          1671        38
                C-----TOP                                                                 1672
00150EI                 YP=YN(III)                                                        1673        39
001522I                 YP1=YN1(III)                                                      1674        40
001536I                 CDK=DK(I,J,K)                                                     1675        41
001562I                 CDE=DE(I,J,K)                                                     1676        42
00158EI                 CDEN=DEN(I,J,K)                                                   1677        43
0015BAI                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1678        44
                       1      SINZ(III),F(I,J,K+1),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1679
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1680
                       3      U(I,J,K+1),V(I,J,K+1),W(I,J,K+1),U(I,J,K),V(I,J,K),       1681
                       4      W(I,J,K),U(I,J,K-1),V(I,J,K-1),W(I,J,K-1),                1682
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1683
001A16I                 GO TO 150                                                         1684        45
001A1EI           6     CONTINUE                                                          1685        46
                C-----BOTTOM                                                              1686
001A1EI                 YP=YN(III)                                                        1687        47
001A32I                 YP1=YN1(III)                                                      1688        48
001A46I                 CDK=DK(I,J,K)                                                     1689        49
001A72I                 CDE=DE(I,J,K)                                                     1690        50
001A9EI                 CDEN=DEN(I,J,K)                                                   1691        51
001ACAI                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1692        52
                       1      SINZ(III),F(I,J,K-1),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1693
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1694
                       3      U(I,J,K-1),V(I,J,K-1),W(I,J,K-1),U(I,J,K),V(I,J,K),       1695
                       4      W(I,J,K),U(I,J,K+1),V(I,J,K+1),W(I,J,K+1),                1696
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1697
001F28I          150  CONTINUE                                                          1698        53
001F40I                 RETURN                                                            1699        54
001F46I                 END                                                               1700        55
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NO  ERRORS:F7D  R05-01.DC  SUBROUTINE  BOUNC      02/21/86  10:00:46  TABLE SPACE:    7 KB
STATEMENT BUFFER:    20 LINES/1321 BYTES     STACK SPACE: 136 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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      SUBROUTINE WALLFN(IE,YP,YP1,DK1,DE1,DEN1,SINX1,SINY1,                1701
     1      SINZ1,F1,AN1,SU1,SP1,                                          1702
     2      TAUN1,YPLN1,GEN1,VISE1,                                        1703
     3      U1,V1,W1,UC,VC,                                                1704
     4      WC,U2,V2,W2,                                                   1705
     5      SUK1,SPK1,ANV2,ANW2,TJC1)                                      1706
      COMMON                                                               1707
     1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                1708
     1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,MINUM,SMNUM,ANV1(800),      1709
     2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),           1710
     3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(300),           1711
     4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO            1712
C-----WALL FUNCTIONS USING LOGARITHMIC WALL LAW                            1713
      GO TO (1,2,3,4,5,6,7), IE                                            1714      1
    1 CONTINUE                                                             1715      2
C-----U                                                                    1716
      SQRTK=SQRT(DK1)                                                      1717      3
      YPLN1=DEN1*SQRTK*CMU1*YP/VISC                                        1718      4
      IF(YPLN1 .LE. 11.63) GO TO 111                                       1719      5
      TMULT=DEN1*CMU1*SQRTK*CK/ALOG(E*YPLN1)                               1720      7
      GO TO 112                                                            1721      8
  111 TMULT=VISC/YP                                                        1722      9
  112 TAUN1=-TMULT                                                         1723     10
      PTA=TJC1*SINX1*TMULT/YP1                                             1724     11
      SP1=SP1-PTA                                                          1725     12
      SU1=SU1+PTA*F1                                                       1726     13
      P1=SQRT(1.-SINX1**2)                                                 1727     14
      ANV2=AN1*SQRT(1.-SINY1**2)                                           1728     15
      ANW2=AN1*SQRT(1.-SINZ1**2)                                           1729     16
      AN1=P1*AN1                                                           1730     17
      RETURN                                                               1731     18
    2 CONTINUE                                                             1732     19
C-----V                                                                    1733
      TMULT=-TAUN1                                                         1734     20
      PTA=TJC1*SINY1*TMULT/YP1                                             1735     21
      SP1=SP1-PTA                                                          1736     22
      SU1=SU1+PTA*F1                                                       1737     23
      AN1=ANV2                                                             1738     24
      RETURN                                                               1739     25
    3 CONTINUE                                                             1740     26
C-----W                                                                    1741
      TMULT=-TAUN1                                                         1742     27
      PTA=TJC1*SINZ1*TMULT/YP1                                             1743     28
      SP1=SP1-PTA                                                          1744     29
      SU1=SU1+PTA*F1                                                       1745     30
      AN1=ANW2                                                             1746     31
      RETURN                                                               1747     32
    4 CONTINUE                                                             1748     33
C-----TM                                                                   1749
      RETURN                                                               1750     34
    5 CONTINUE                                                             1751     35
C-----K                                                                    1752
      SQRTK=SQRT(DK1)                                                      1753     36
      IF(YPLN1 .LE. 11.63) GO TO 511                                       1754     37
      DITM=DEN1*CMU2*SQRTK*ALOG(E*YPLN1)/(CK*YP)                           1755     39
      GO TO 512                                                            1756     40
  511 DITM=DEN1*CMU2*SQRTK*YPLN1/YP                                        1757     41
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 58  0003ACI     512  CONTINUE                                                                1758      42
 59  0003ACI          DDU=U0-U1                                                               1759      43
 60  00038EI          DDV=V0-V1                                                               1760      44
 61  000300I          DDW=W0-W1                                                               1761      45
 62  0003E2I          P1=DDU**2+DDV**2+DDW**2                                                 1762      46
 63  00042AI          DDU=U2-U1                                                               1763      47
 64  00043CI          DDV=V2-V1                                                               1764      48
 65  00044EI          DDW=W2-W1                                                               1765      49
 66  000460I          P2=DDU**2+DDV**2+DDW**2                                                 1766      50
 67  0004A8I          YP2=2*YP1-YP                                                            1767      51
 68  0004C0I          GENR=GEN1-VISE1*P2/YP2/YP2                                              1768      52
 69  0004E6I          IF(GENR .LE. 0.0) GENR=0.0                                              1769      53
 70  000504I          SU1=TJC1*(SUK1*DK1*(P1*TAUN1**2/VISE1+GENR))                            1770      55
 71  00053EI          SP1=TJC1*(SPK1-DITM)                                                    1771      56
 72  000556I          AN1=0.0                                                                 1772      57
 73  000562I          RETURN                                                                  1773      58
 74  000568I        6 CONTINUE                                                                1774      59
 75               C-----E                                                                   1775
 76  000568I          TERM=CMU2/(CK*YP)                                                       1776      60
 77  000582I          SU1=HINUM*TERM*DK1**1.5                                                 1777      61
 78  000532I          SP1=-HINUM                                                              1778      62
 79  0005C4I          RETURN                                                                  1779      63
 80  0005CAI        7 CONTINUE                                                                1780      64
 81               C-----PF                                                                  1781
 82  0005CAI          RETURN                                                                  1782      65
 83  0005D0I          END                                                                     1783      66

NO  ERRORS:F7D  R05-01.00  SUBROUTINE  WALLFN    02/21/86  10:01:12  TABLE SPACE:    5 KB
STATEMENT BUFFER:    20 LINES/1321 BYTES    STACK SPACE: 122 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
 1  000000I          SUBROUTINE SYMOUT(IC,IE,IS,IT,JS,JT,KS,KT)                                  1784
 2  000004I          COMMON                                                                      1785
 3                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),                     1786
 4                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                              1787
 5                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                    1788
 6                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                      1789
 7                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                              1790
 8                  1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),               1791
 9                  2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),                  1792
10                  3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)                    1793
11  000004I          COMMON                                                                      1794
12                  1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),                  1795
13                  2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                                 1796
14                  3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                                 1797
15                  3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                                  1798
16                  1/UNSTDY/UO(21,18,10),VO(21,18,10),WO(21,18,10),DKO(21,18,10),              1799
17                  2   DEO(21,18,10),DENO(21,18,10),TMO(21,18,10)                              1800
18                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,               1801
19                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT             1802
20              C-----SYMMETRIC, CYCLIC AND EXIT CONDITIONS AND LINK MODIFICATIONS               1803
21  000004I          GO TO (1,2,3), IC                                                           1804        1
22  00002EI      1   CONTINUE                                                                    1805        2
23              C-----BOTTOM                                                                    1806
24  00002EI          K=1                                                                         1807        3
25  000036I          DO 10 I=1,L                                                                 1808        4
26  00004AI          DO 10 J=2,MT                                                                1809        5
27  00005EI          U(I,J,K)=U(I,J,K+1)                                                         1810        6
28  0000ACI          V(I,J,K)=V(I,J,K+1)                                                         1811        7
29  0000FAI          W(I,J,K)=0.0                                                                1812        8
30  00012EI          TM(I,J,K)=TM(I,J,K+1)                                                       1813        9
31  000174I          DK(I,J,K)=DK(I,J,K+1)                                                       1814       10
32  0001C2I          DE(I,J,K)=DE(I,J,K+1)                                                       1815       11
33  000210I      10  CONTINUE                                                                    1816       12
34              C-----EAST OUT (BASED ON INFLOW MASS FLOW RATE)                                  1817
35  000240I          I=IT                                                                        1818       13
36  00024CI          FLOW=0.0                                                                    1819       14
37  000258I          ARDEN=0.0                                                                   1820       15
38  000264I          DO 50 J=2,JT                                                                1821       16
39  000278I          DO 50 K=2,KT                                                                1822       17
40  00028CI          UC=(V(I,J,K)+V(I,J-1,K)+V(I,J,K-1)+V(I,J-1,K-1))*0.25                       1823       18
41  00033CI          DENC=(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K-1)+DEN(I,J-1,K-1))*0.25             1824       19
42  0003ECI          P1=(X(I,J,K)+X(I,J,K-1)-X(I,J-1,K)-X(I,J-1,K-1))*0.5                        1825       20
43  00049CI          P2=(Y(I,J,K)+Y(I,J,K-1)-Y(I,J-1,K)-Y(I,J-1,K-1))*0.5                        1826       21
44  00054CI          P3=(Z(I,J,K)+Z(I,J,K-1)-Z(I,J-1,K)-Z(I,J-1,K-1))*0.5                        1827       22
45  0005FCI          Q1=(X(I,J,K)+X(I,J-1,K)-X(I,J,K-1)-X(I,J-1,K-1))*0.5                        1828       23
46  0006ACI          Q2=(Y(I,J,K)+Y(I,J-1,K)-Y(I,J,K-1)-Y(I,J-1,K-1))*0.5                        1829       24
47  00075CI          Q3=(Z(I,J,K)+Z(I,J-1,K)-Z(I,J,K-1)-Z(I,J-1,K-1))*0.5                        1830       25
48  00080CI          AREA=SQRT(P1*P1+P2*P2+P3*P3)*SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                         1831       26
49  0008A8I          FLOW=FLOW+DENC*AREA*UC                                                      1832       27
50  0008C6I          ARDEN=ARDEN+DENC*AREA                                                       1833       28
51  0008DEI      50  CONTINUE                                                                    1834       29
52  0009DEI          UINC=(FLOW-FLOWIN)/ARDEN                                                    1835       30
53  000926I          DO 60 J=2,JT                                                                1836       31
54  00093AI          DO 60 K=2,KT                                                                1837       32
55  00094EI          U(I+1,J,K)=U(I,J,K)                                                         1838       33
56  00099EI          V(I+1,J,K)=V(I,J,K)-UINC                                                    1839       34
57  0009F4I          W(I+1,J,K)=W(I,J,K)                                                         1840       35
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 58  000A44I          DK(I+1,J,K)=DK(I,J,K)                                  1841      36
 59  000A94I          DE(I+1,J,K)=DE(I,J,K)                                  1842      37
 60  000AE4I       60 CONTINUE                                               1843      38
 61  000B14I          RETURN                                                 1844      39
 62                C-----LINK COEFF. MODIFICATIONS                           1845
 63  000B1AI       2  CONTINUE                                               1846      40
 64                C-----EAST OUT                                            1847
 65  000B1AI          I=IT                                                   1848      41
 66  000B26I          DO 200 J=2,JT                                          1849      42
 67  000B3AI          DO 200 K=2,KT                                          1850      43
 68  000B4EI          AE(I,J,K)=0.0                                          1851      44
 69  000B7AI     200  CONTINUE                                               1852      45
 70  000BAAI          RETURN                                                 1853      46
 71                C-----UPDATE UNSTEADY COEFF.                              1854
 72  000BB0I       3  CONTINUE                                               1855      47
 73  000BB0I          IF(DTT .NE. 0.0) GO TO 301                             1856      48
 74  000BC8I          DO 300 I=IS,IT                                         1857      50
 75  000BE0I          DO 300 J=JS,JT                                         1858      51
 76  000BF8I          DO 300 K=KS,KT                                         1859      52
 77  000C10I     300  APO(I,J,K)=0.0                                         1860      53
 78  000C84I          RETURN                                                 1861      54
 79  000C8AI     301  CONTINUE                                               1862      55
 80  000C8AI          DO 310 I=IS,IT                                         1863      56
 81  000CA2I          DO 310 J=JS,JT                                         1864      57
 82  000CBAI          DO 310 K=KS,KT                                         1865      58
 83  000CD2I          APO(I,J,K)=DENO(I,J,K)/DTT                             1866      59
 84  000D24I          UO(I,J,K)=U(I,J,K)                                     1867      60
 85  000D70I          VO(I,J,K)=V(I,J,K)                                     1868      61
 86  000DBCI          WO(I,J,K)=W(I,J,K)                                     1869      62
 87  000E08I          TMO(I,J,K)=TM(I,J,K)                                   1870      63
 88  000E54I          DKO(I,J,K)=DK(I,J,K)                                   1871      64
 89  000EA0I          DEO(I,J,K)=DE(I,J,K)                                   1872      65
 90  000EECI     310  CONTINUE                                               1873      66
 91  000F34I          RETURN                                                 1874      67
 92  000F3AI          END                                                    1875      68

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  SYMOUT    02/21/86  10:01:59  TABLE SPACE:   7 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 181 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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ORIGINAL PAGE IS
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  1  0000001          SUBROUTINE WALVAL(PW,IS,IT,JS,JT,KS,KT,F)                        1876
  2  0000041          DIMENSION F(21,18,10)                                            1877
  3  0000341          COMMON                                                           1878
  4                  1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),  1879
  5                  2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),      1880
  6                  3 YPLN(800),TAUN(800),I3C(800),J3C(800),K3C(800),IITY(800),      1881
  7                  4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO       1882
  8              C-----ASSIGN WALL VALUES                                             1883
  9  0000941          DO 10 J=JS,JT                                                    1884      1
 10  0000281          DO 10 K=KS,KT                                                    1885      2
 11  0000401          F(IS-1,J,K)=PW*F(IS,J,K)                                         1886      3
 12  0000A21     10   F(IT+1,J,K)=PW*F(IT,J,K)                                         1887      4
 13  0001341          DO 20 I=IS-1,IT+1                                                1888      5
 14  0001521          DO 20 K=KS,KT                                                    1889      6
 15  00016A1          F(I,JS-1,K)=PW*F(I,JS,K)                                         1890      7
 16  0001CC1     20   F(I,JT+1,K)=PW*F(I,JT,K)                                         1891      8
 17  00025E1          DO 30 I=IS-1,IT+1                                                1892      9
 18  0002701          DO 30 J=JS-1,JT+1                                                1893     10
 19  00029A1          F(I,J,KS-1)=PW*F(I,J,KS)                                         1894     11
 20  0002FA1     30   F(I,J,KT+1)=PW*F(I,J,KT)                                         1895     12
 21  00038A1          DO 40 III=1,IITO                                                 1896     13
 22  00039E1          I=I3C(III)                                                       1897     14
 23  0003B21          J=J3C(III)                                                       1898     15
 24  0003C61          K=K3C(III)                                                       1899     16
 25  0003DA1          GO TO (1,2,3,4,5,6), IITY(III)                                   1900     17
 26  0004161     1    F(I,J+1,K)=PW*F(I,J,K)                                           1901     18
 27  0004781          GO TO 40                                                         1902     19
 28  0004781     2    F(I,J-1,K)=PW*F(I,J,K)                                           1903     20
 29  0004E01          GO TO 40                                                         1904     21
 30  0004E61     3    F(I+1,J,K)=PW*F(I,J,K)                                           1905     22
 31  0005481          GO TO 40                                                         1906     23
 32  00054E1     4    F(I-1,J,K)=PW*F(I,J,K)                                           1907     24
 33  0005801          GO TO 40                                                         1908     25
 34  0005861     5    F(I,J,K+1)=PW*F(I,J,K)                                           1909     26
 35  0006161          GO TO 40                                                         1910     27
 36  00061C1     6    F(I,J,K-1)=PW*F(I,J,K)                                           1911     28
 37  00067C1     40   CONTINUE                                                         1912     29
 38  0006941          RETURN                                                           1913     30
 39  00069A1          END                                                              1914     31

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  WALVAL    02/21/86  10:02:24  TABLE SPACE:   3 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 154 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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